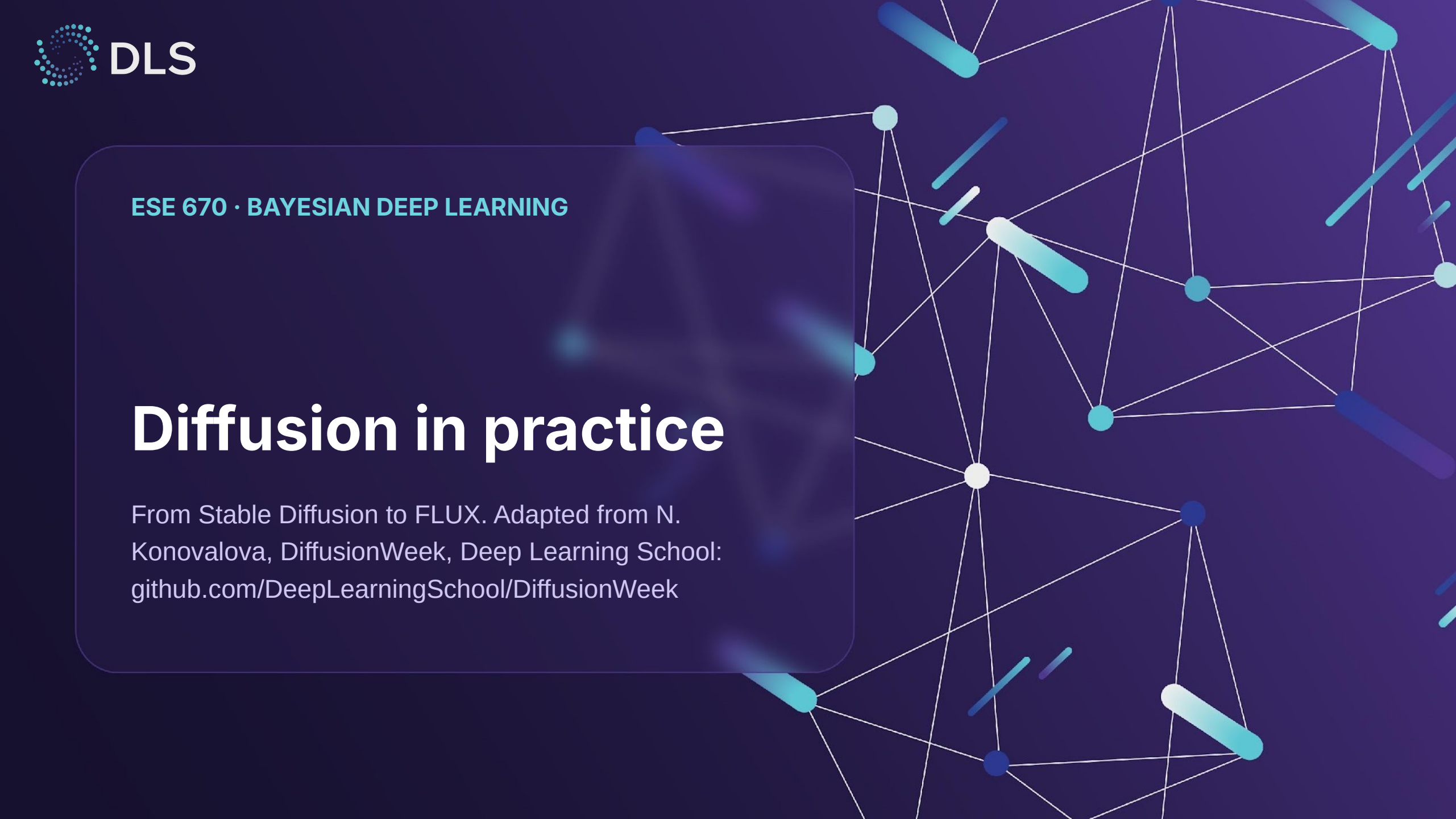
DLS

ESE 670 · BAYESIAN DEEP LEARNING

Diffusion in practice

From Stable Diffusion to FLUX. Adapted from N. Konovalova, DiffusionWeek, Deep Learning School: github.com/DeepLearningSchool/DiffusionWeek



DLS

CREDITS

Thank you

Slides adapted from Nina Konovalova, “From Stable Diffusion to FLUX”

https://github.com/DeepLearningSchool/DiffusionWeek/tree/main/presentations/01_from_sd_to_flux

DLS

00

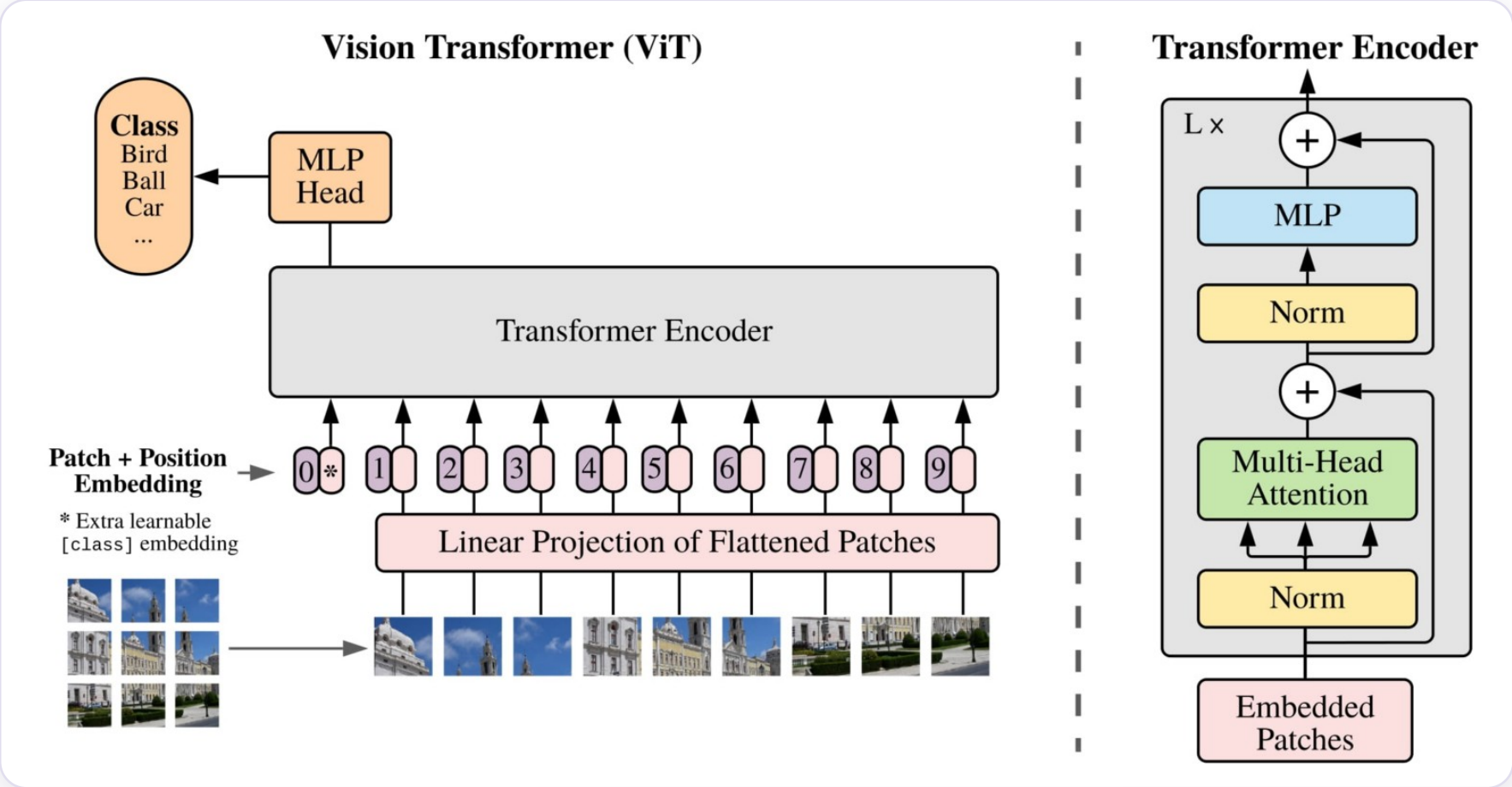
Building blocks

Vision Transformer (ViT) and CLIP



paper →

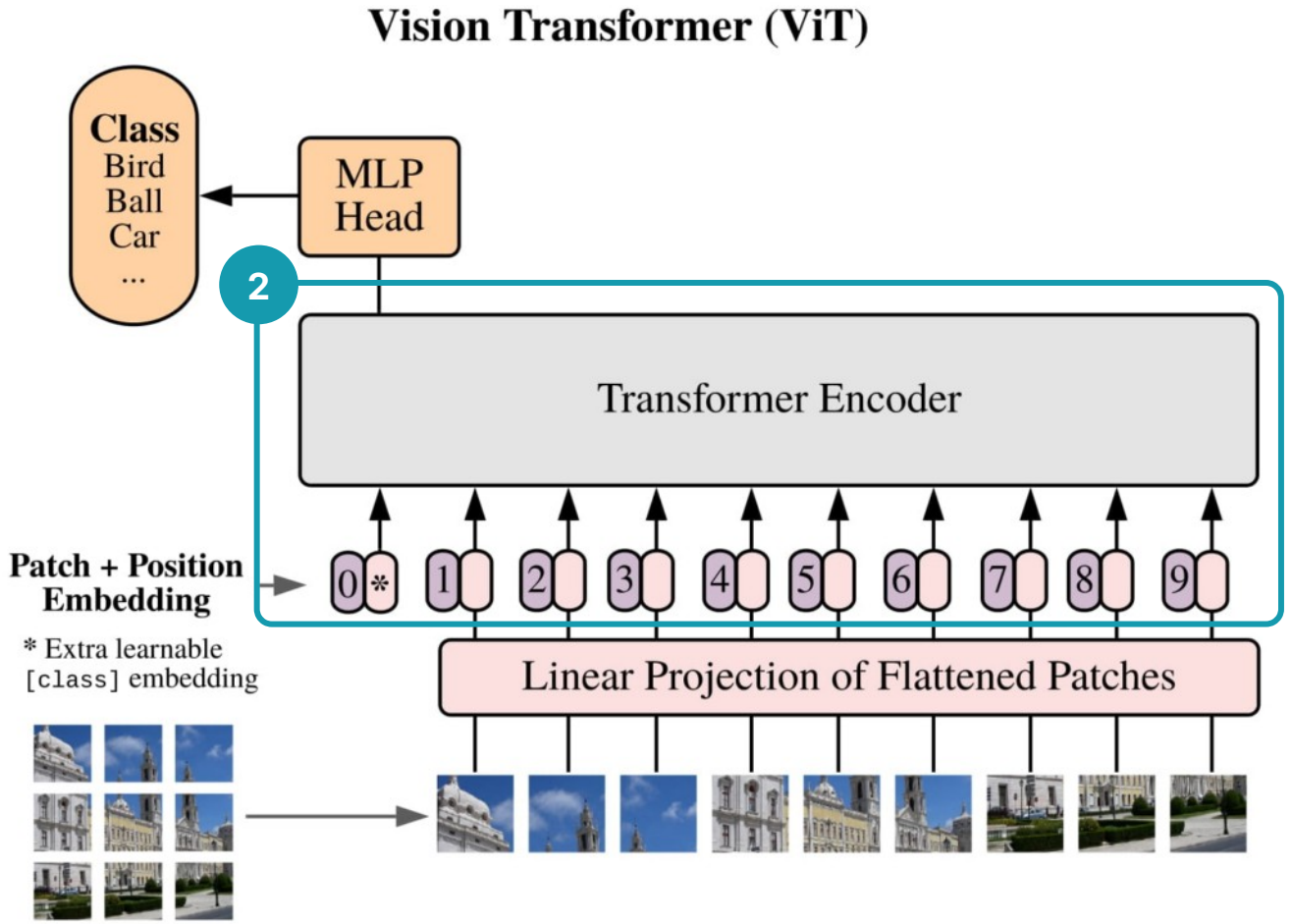
Vision Transformer (ViT)



An image is a sequence of patches: the same transformer encoder as in NLP, with no convolutions

paper →

Vision Transformer (ViT)



1 Patchify

The image is split into fixed-size patches (e.g. 16×16). Each patch is flattened, linearly projected to a token and gets a position embedding.

2 Transformer encoder

Image tokens are processed as an ordinary sequence by standard transformer blocks. An extra [class] token is used for classification.

CLIP: contrastive language–image pre-training

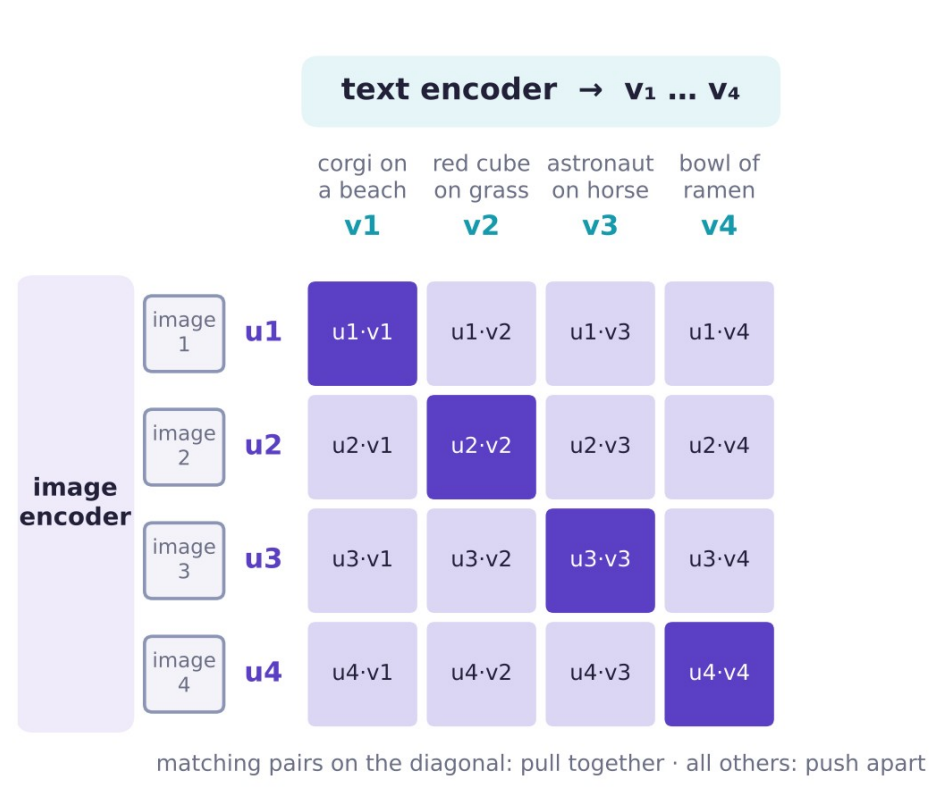
Two encoders, one space — an image encoder (ViT or ResNet) and a text encoder (Transformer) map an image and a caption to unit vectors u and v.

Contrastive training — a batch of N image–caption pairs gives an N × N matrix of similarities ⟨u_i, v_j⟩ / T. A symmetric cross-entropy asks each image to pick its own caption among N, and each caption its image.

Data — 400M web image–caption pairs (OpenAI CLIP); OpenCLIP is trained on LAION-2B.

Zero-shot classification — compare an image with “a photo of a {class}” for every class.

In this lecture — the text encoder of SD 1.x, SDXL, SD3 and FLUX.1; image embeddings in DALL·E 2; CLIP score as a metric.



CLIP text encoder: what diffusion models use

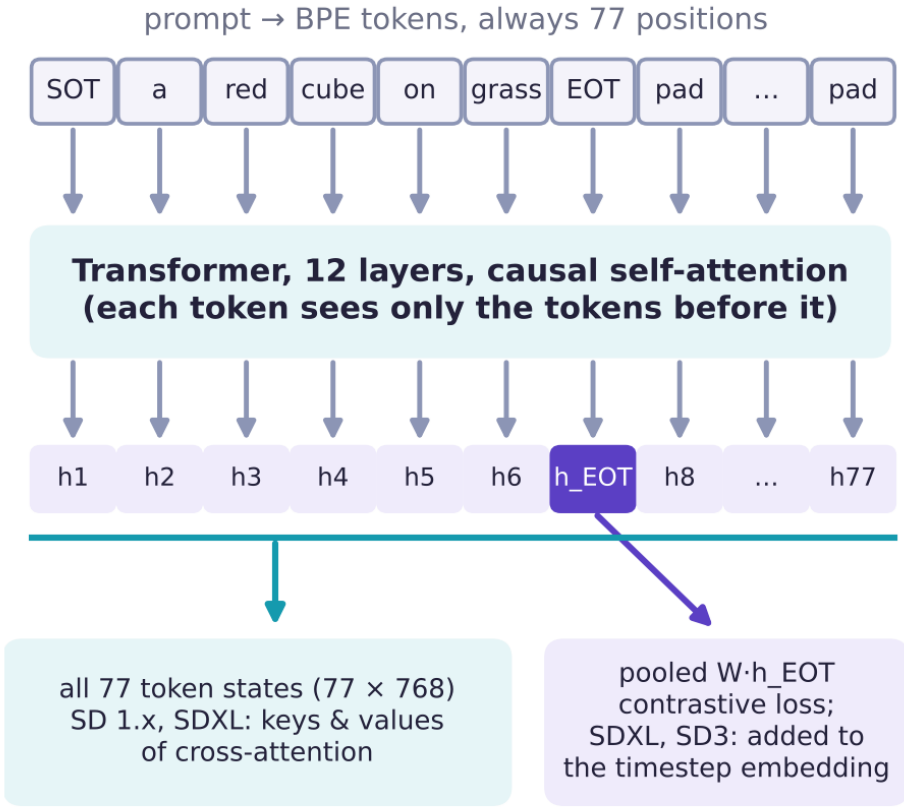
Tokens — BPE with a 49,152-token vocabulary; every prompt is padded or cut to 77 positions.

Causal Transformer — 12 layers, width 768 (ViT-L/14); each position sees only the tokens before it.

Two outputs — the per-token hidden states $h_1 \ldots h_{77}$ (77 × 768) and one pooled vector W·h_EOT. Only the pooled vector enters the contrastive loss.

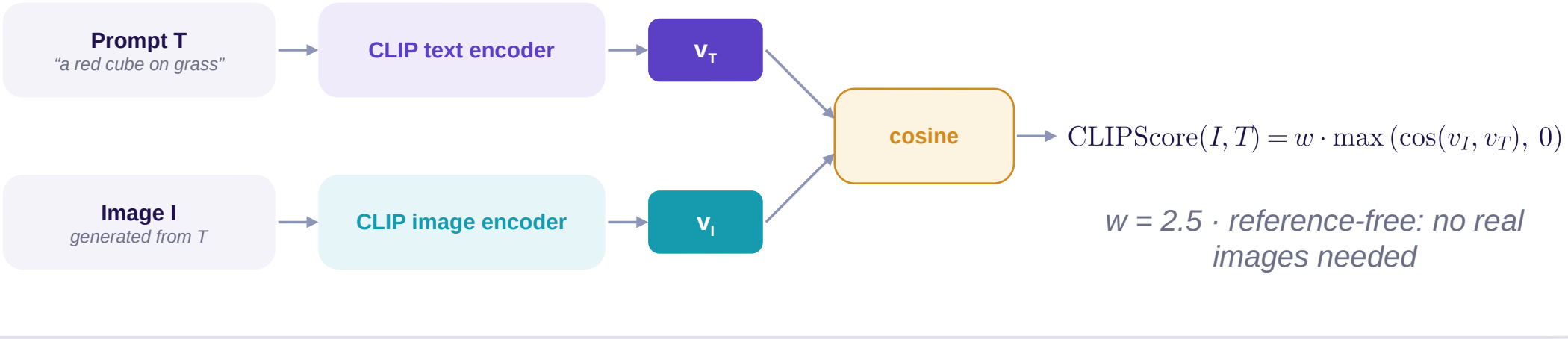
What diffusion uses — SD 1.x takes all 77 token states as keys and values of cross-attention, so each pixel can pick its words. SDXL and SD3 also add the pooled vector to the timestep embedding.

Limits — 77-token cap, one-directional context, caption-level training: weak at counting and attribute binding. Imagen and SD3 add T5.



paper →

CLIP score



USED FOR

- prompt alignment in almost every T2I paper
- filtering and re-ranking samples
- quick sanity check during training

PROBLEMS

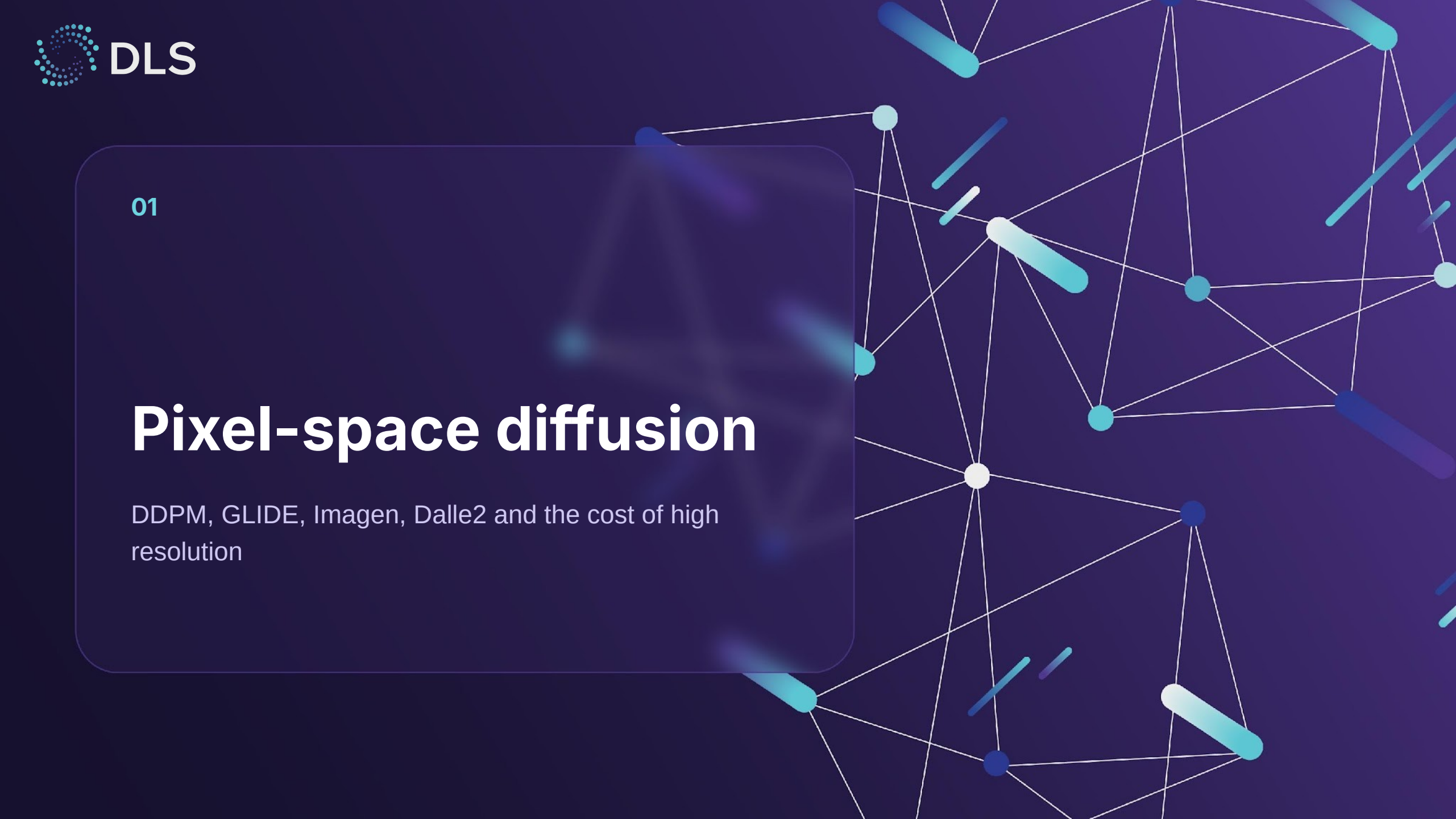
- bag-of-words: ignores word order and relations
- weak at counting, spatial relations, attribute binding
- saturates: strong models get similar scores

DLS

01

Pixel-space diffusion

DDPM, GLIDE, Imagen, Dalle2 and the cost of high resolution



Diffusion models in pixel space



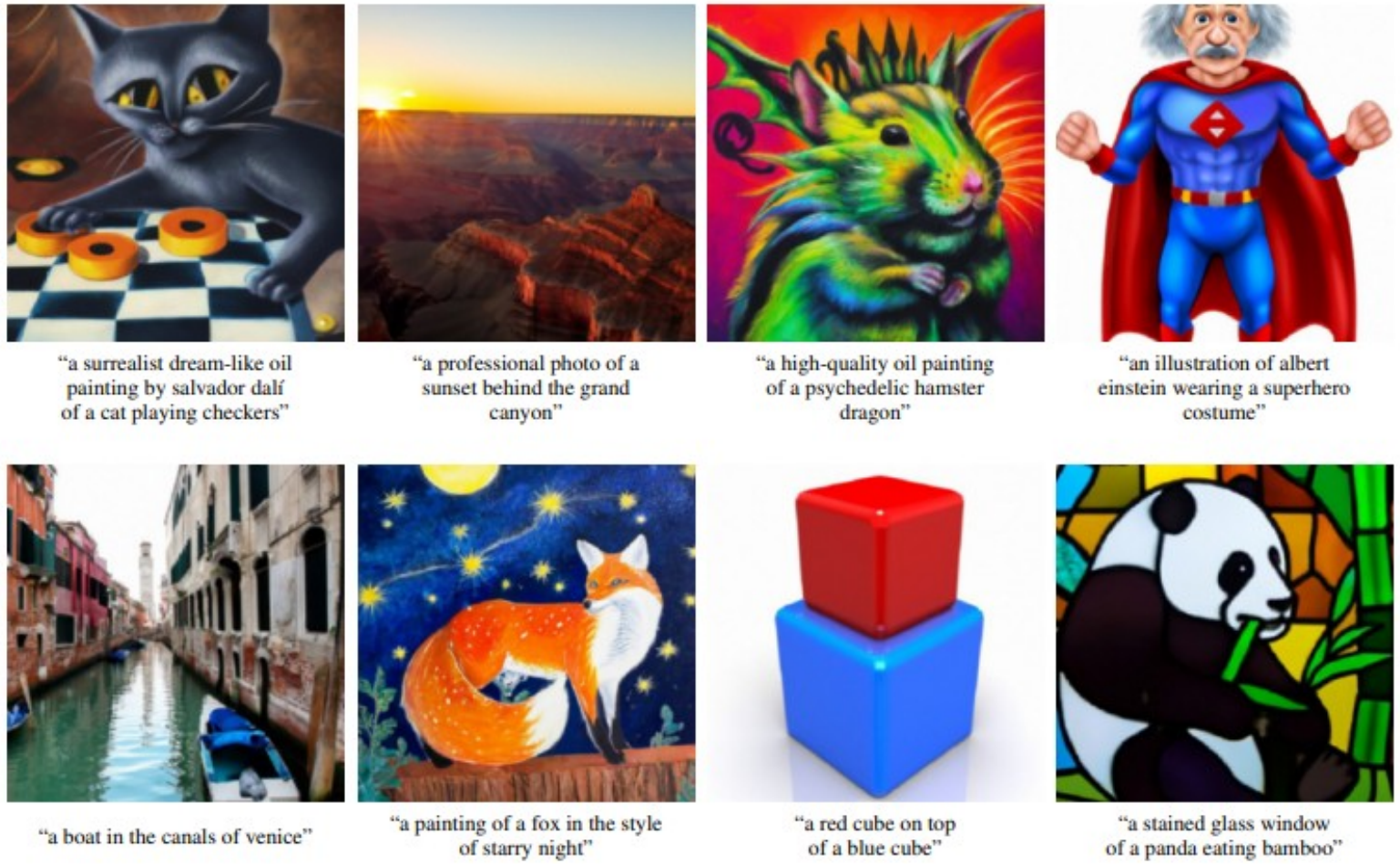
Figure 3: LSUN Church samples. FID=7.89



Figure 4: LSUN Bedroom samples. FID=4.90

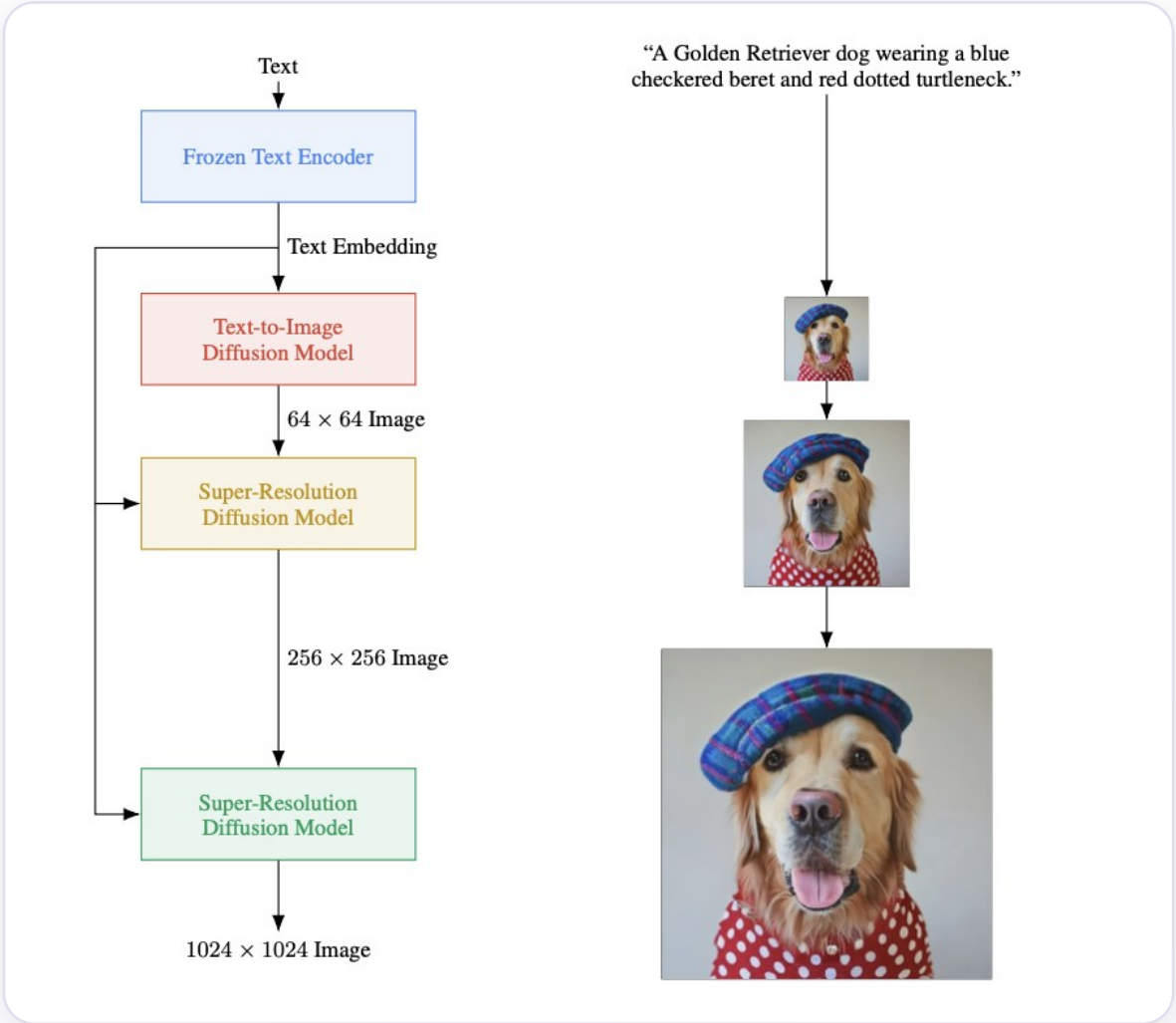
Source

GLIDE: text-to-image generation



Source

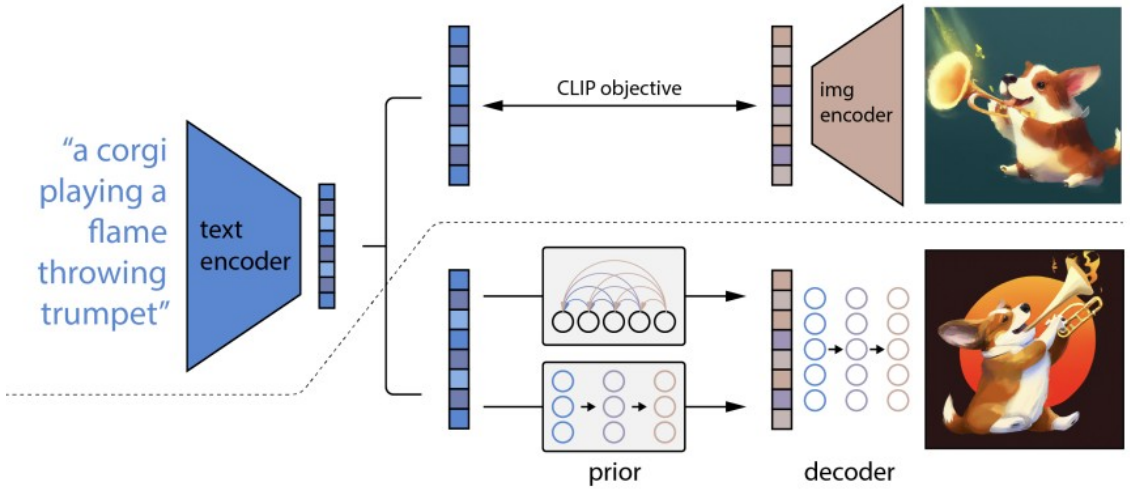
Imagen: a cascade of diffusion models



Source

paper →

DALL·E 2 (unCLIP)

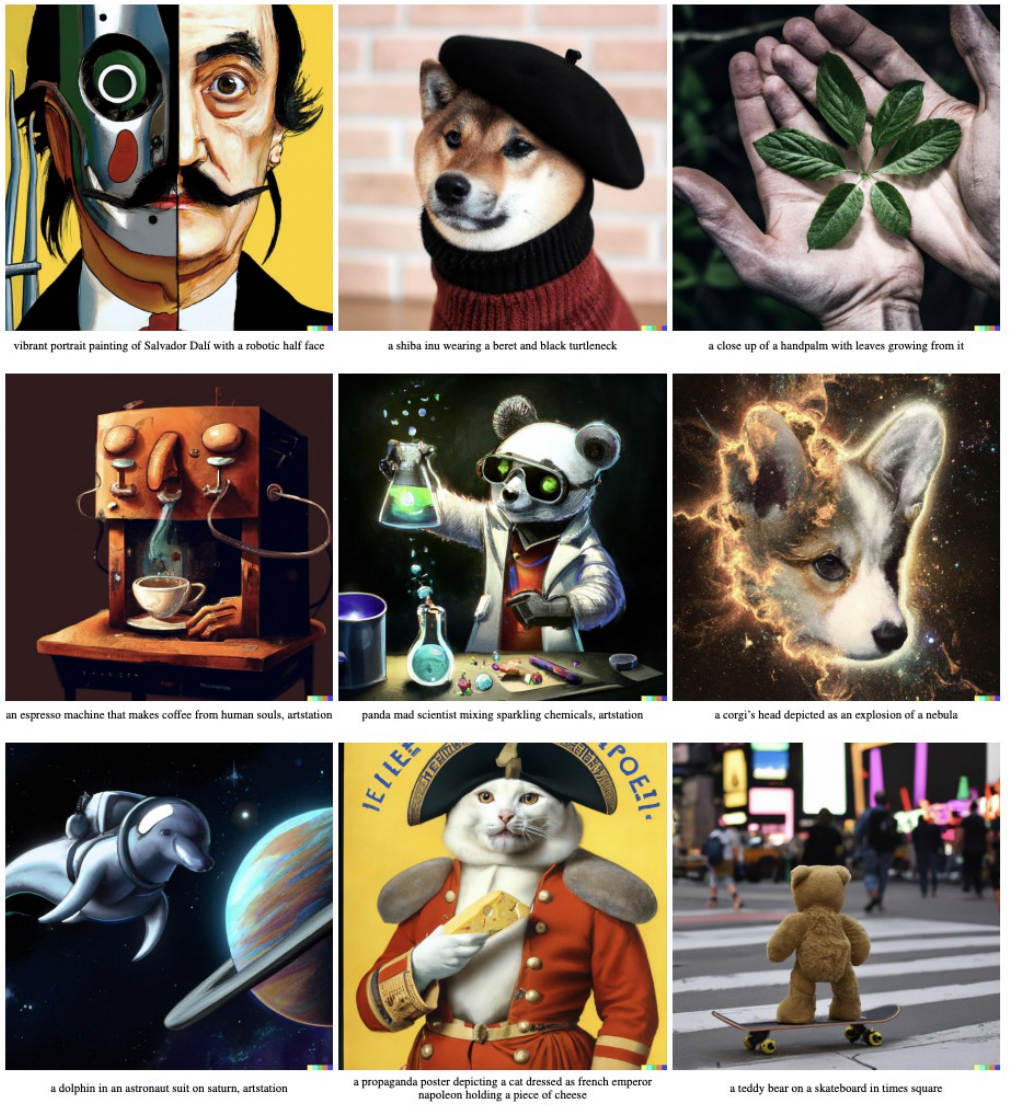


Why a prior?

- Generating the image embedding improves diversity with minimal loss in photorealism
- Guidance only on the decoder

Editing in CLIP space

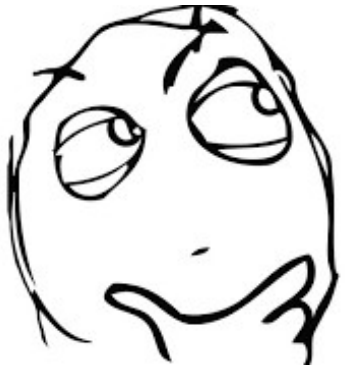
- Variations of an input image
- Interpolations
- Text diffs



Diffusion models in pixel space

Bigger resolution

- More compute for training
- More compute for generation



SPOILER

Sometimes pixel space can work well…

DLS

02

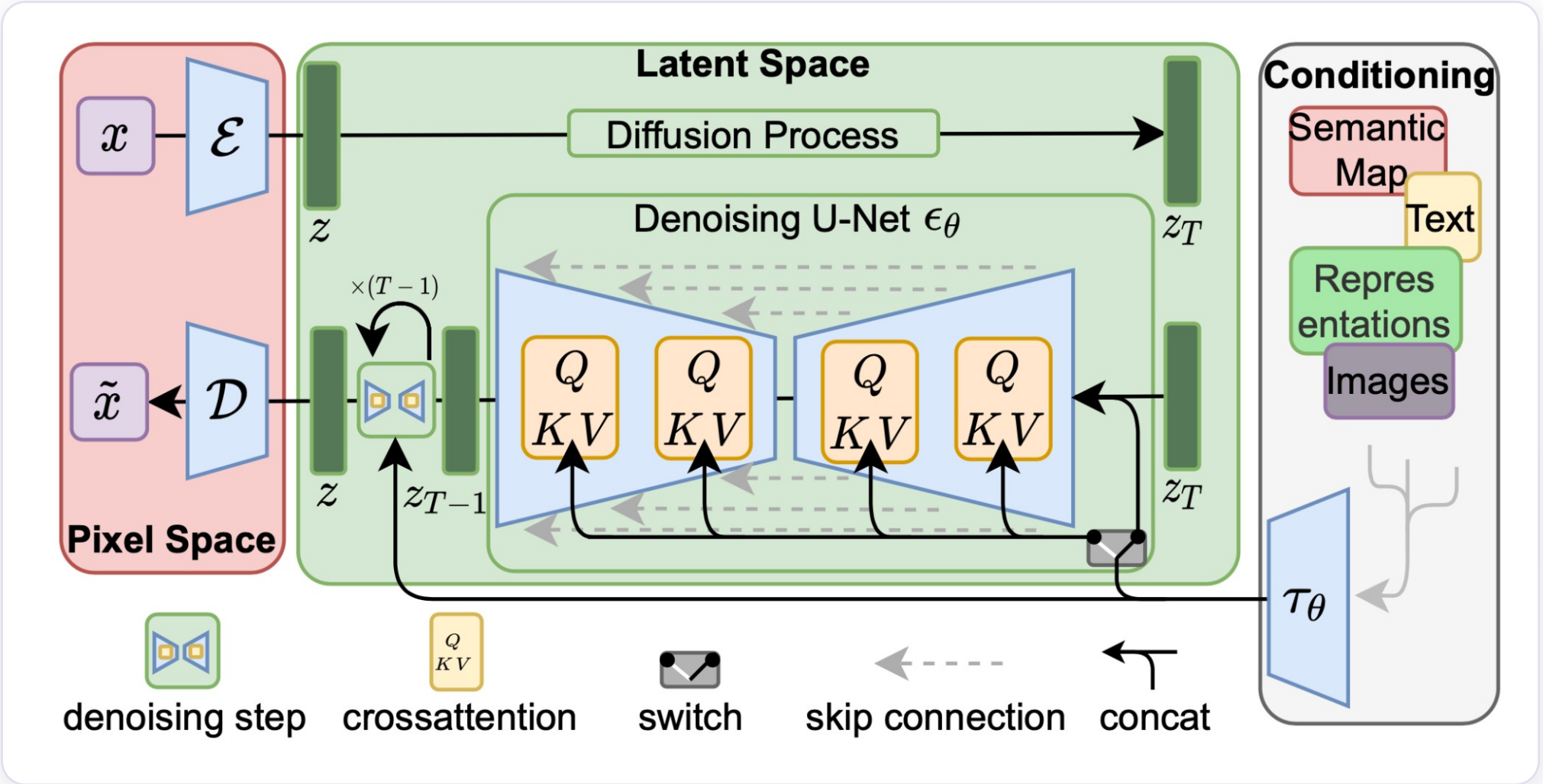
Latent diffusion

Diffusion in the compressed latent space of an autoencoder

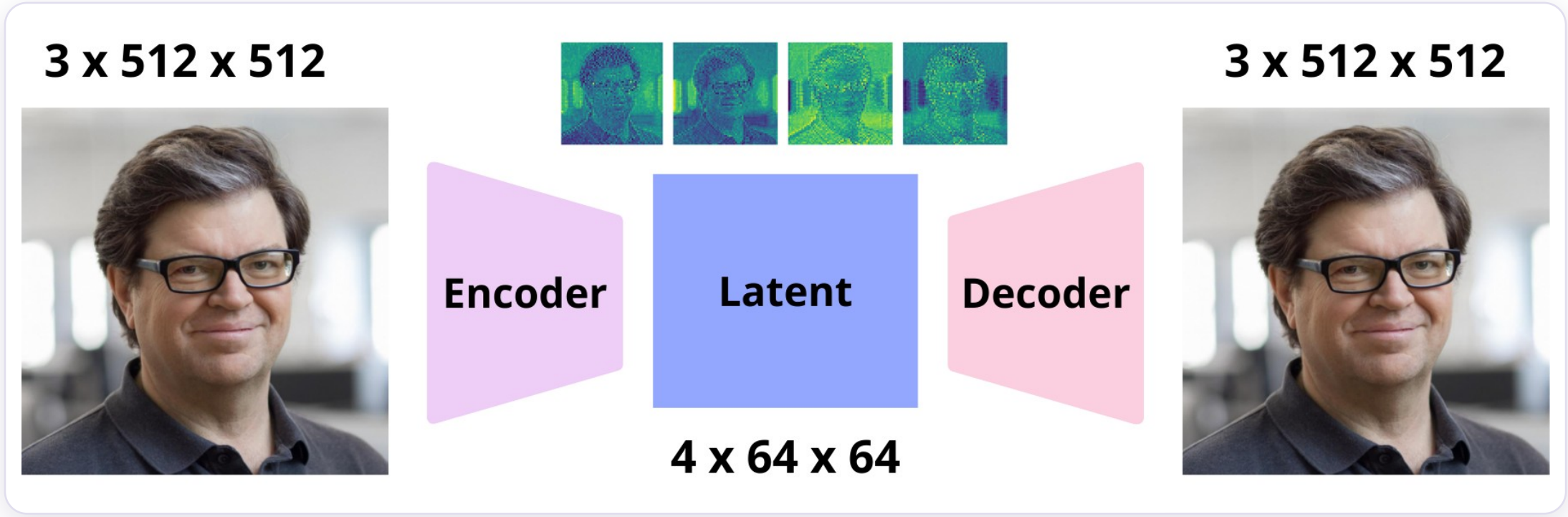


paper →

Latent diffusion

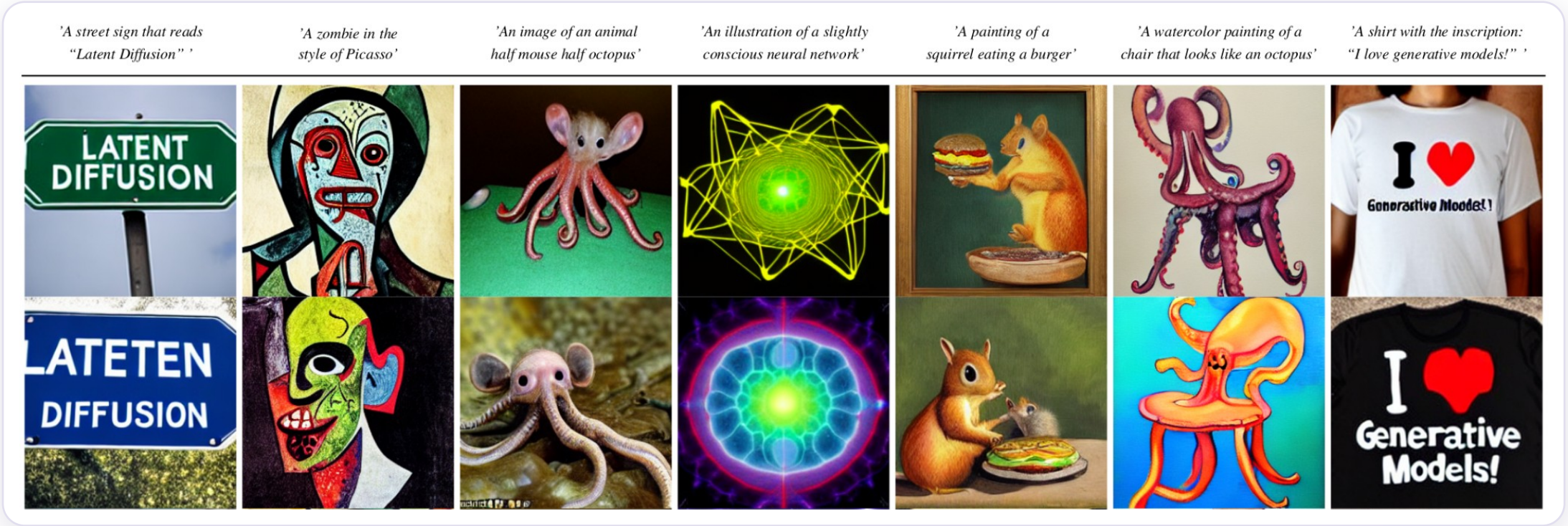


Latent diffusion



paper →

Latent diffusion



DLS

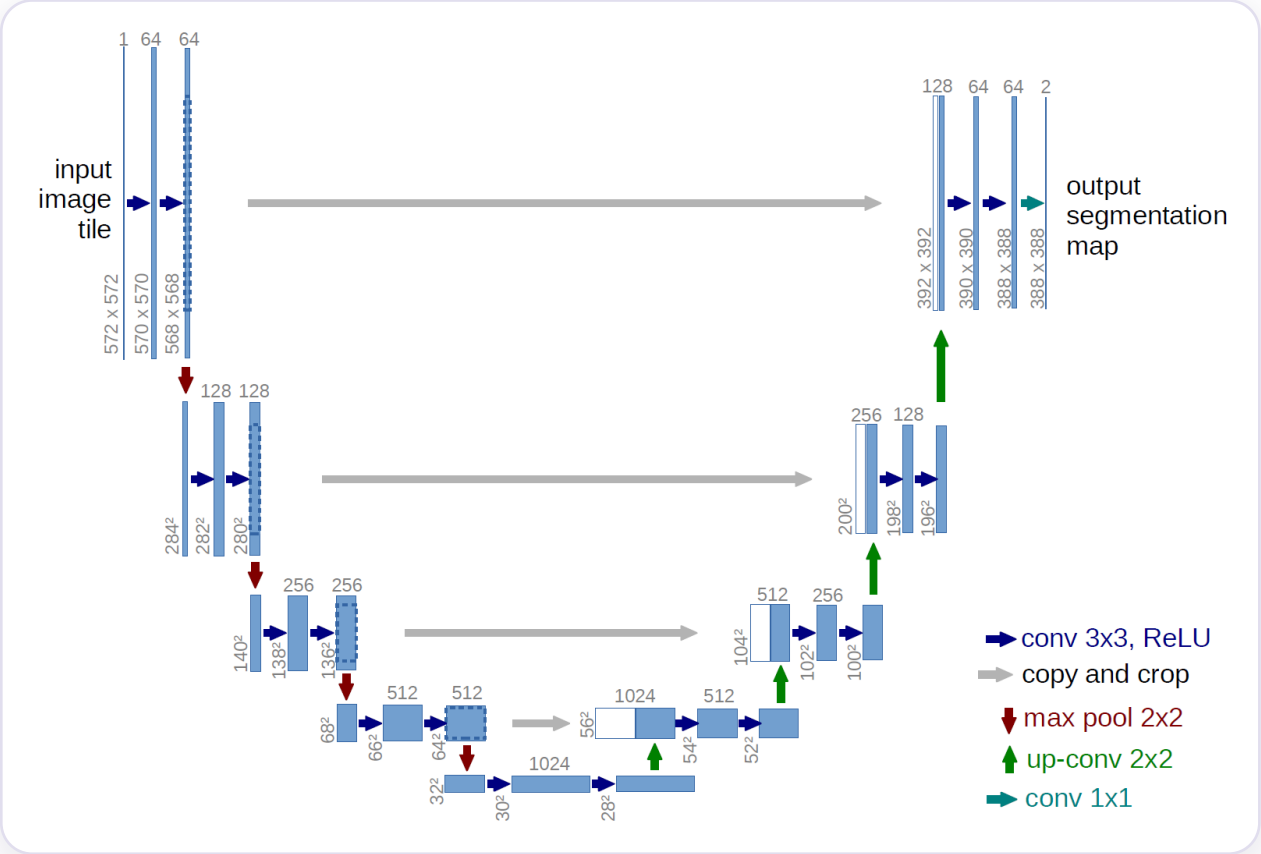
03

UNet text-to-image

Stable Diffusion 1.x, attention maps and SDXL



Text-to-image UNet-based diffusion

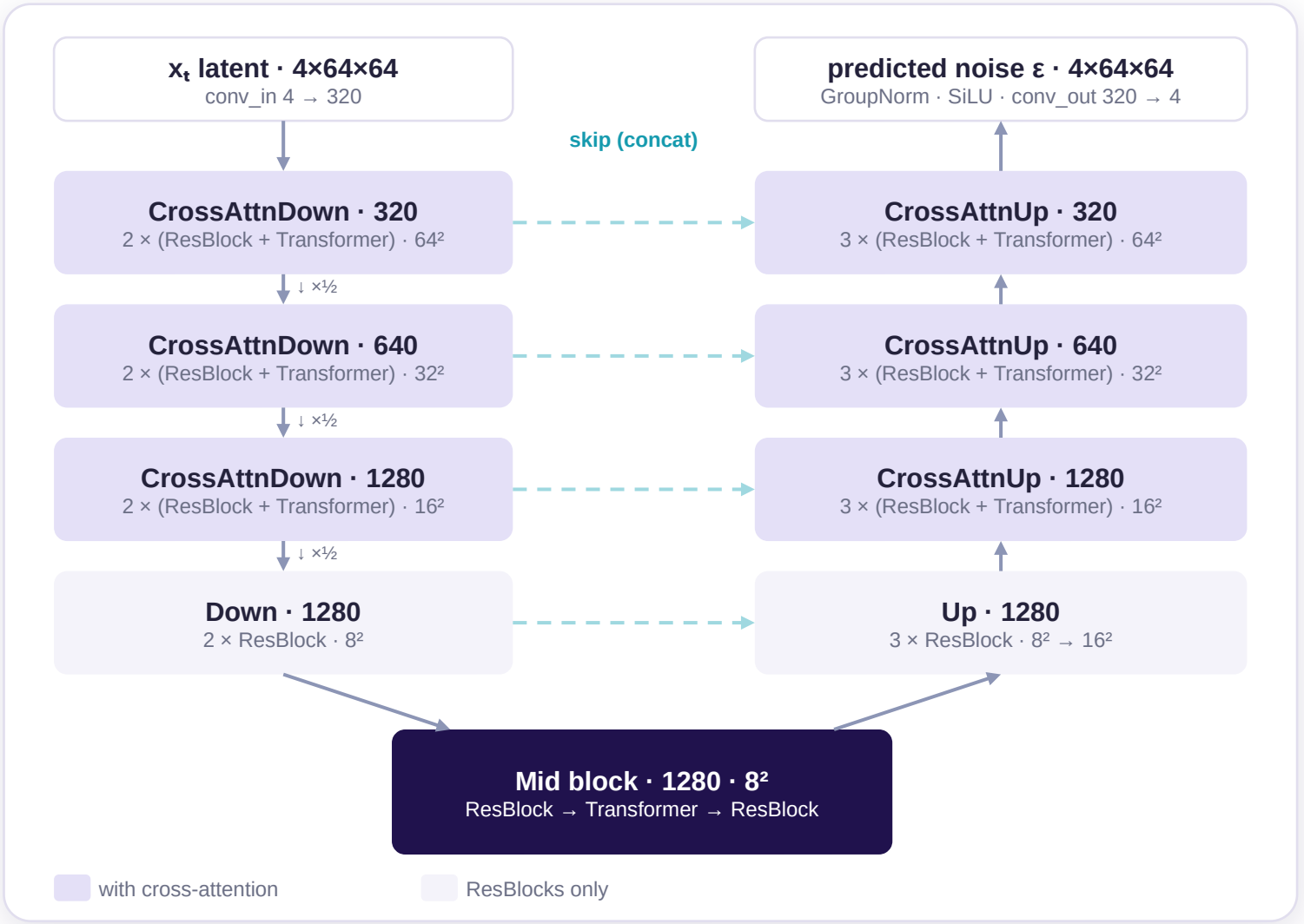


The most popular architecture is old fellow **UNet**, familiar from the segmentation task

UNet is a stack of Residual Convolutional Blocks operating on different resolutions

Deeper layers have more channels and operate on smaller features maps

SD 1.x UNet: overview



TIMESTEP

t → sinusoidal embedding · 320

MLP → temb · 1280

added inside every ResBlock

TEXT PROMPT

CLIP ViT-L/14 text encoder · frozen

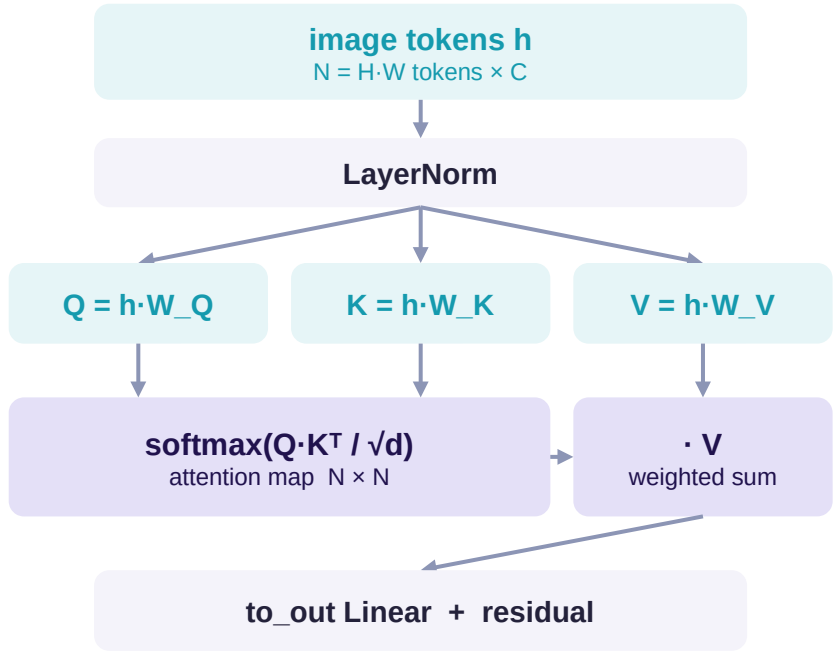
77 tokens × 768

keys & values of every cross-attention

~860M params · 512² image → 64² latent

Self-attention

INSIDE THE SPATIAL TRANSFORMER



$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right)V$$

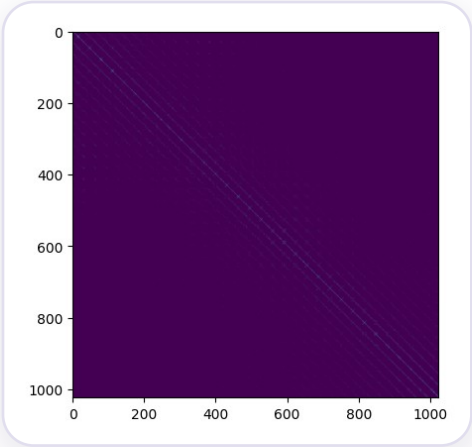
Every pixel can look at every other pixel: global context at the cost of O(N²)



generated image



token grid 32×32 = 1024 tokens



real map: 1024 × 1024

Cross-attention

TEXT ENTERS THE UNET HERE

image tokens h
N = H·W × C

text tokens c
CLIP · 77 × 768

LayerNorm

Q = h·W_Q

K = c·W_K

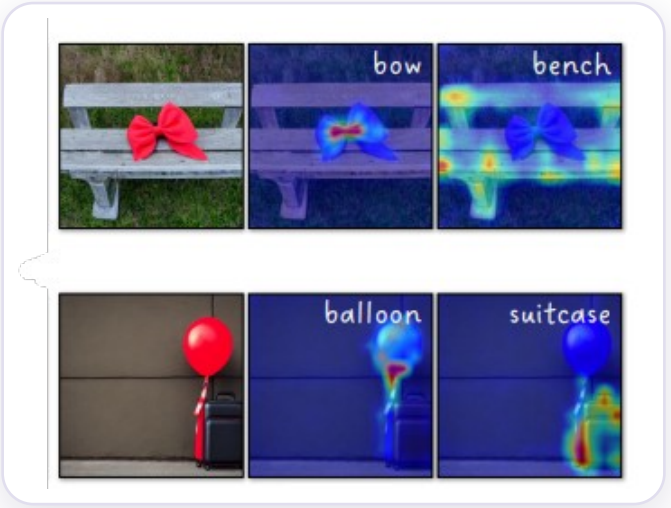
V = c·W_V

softmax(Q·Kᵀ / √d) · V
attention map N × 77 (pixels × words)

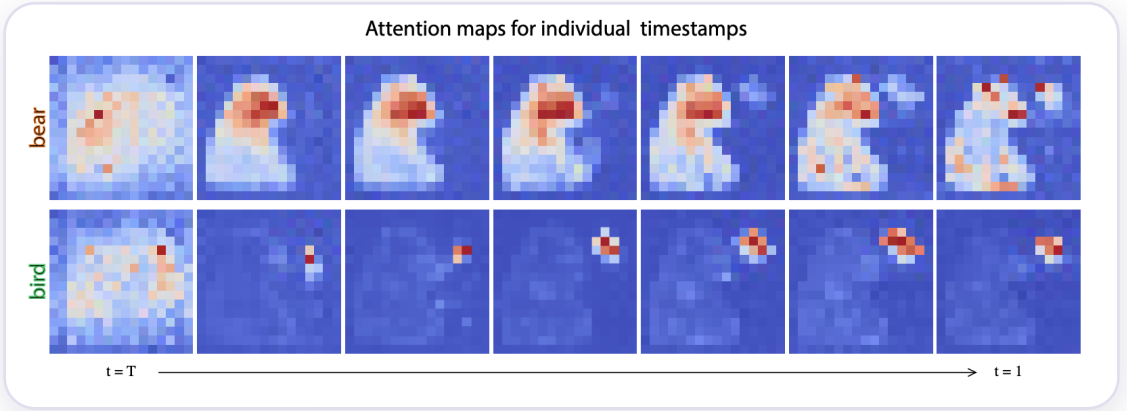
to_out Linear + residual

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right)V$$

Each pixel gathers information from the words; the map for one word shows where it is drawn



cross-attention maps for individual words

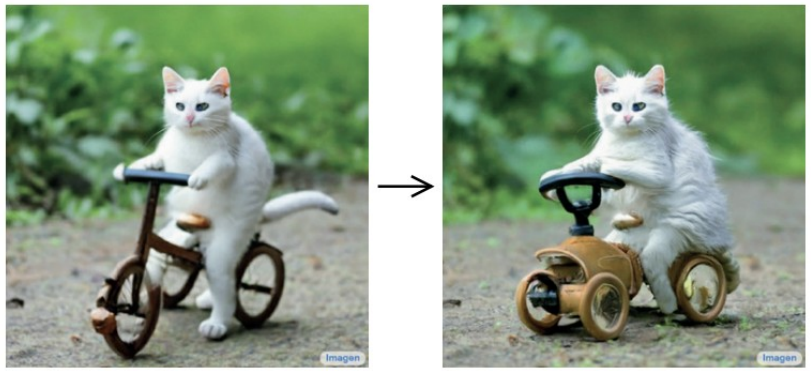


maps for “bear” and “bird” across denoising steps

arxiv.org/pdf/2404.04650 · arxiv.org/pdf/2208.01626

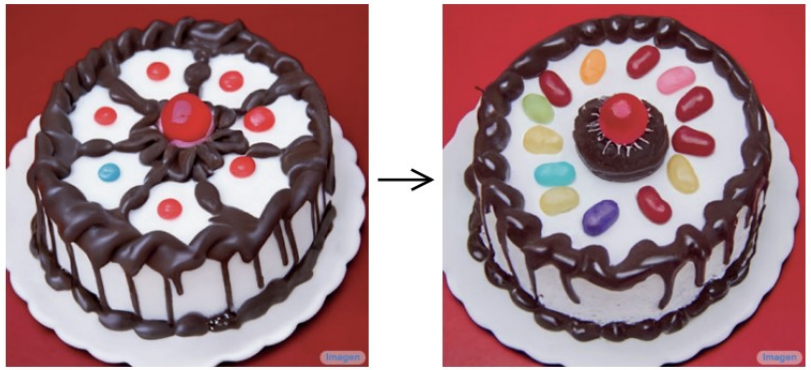
paper →

Prompt-to-prompt



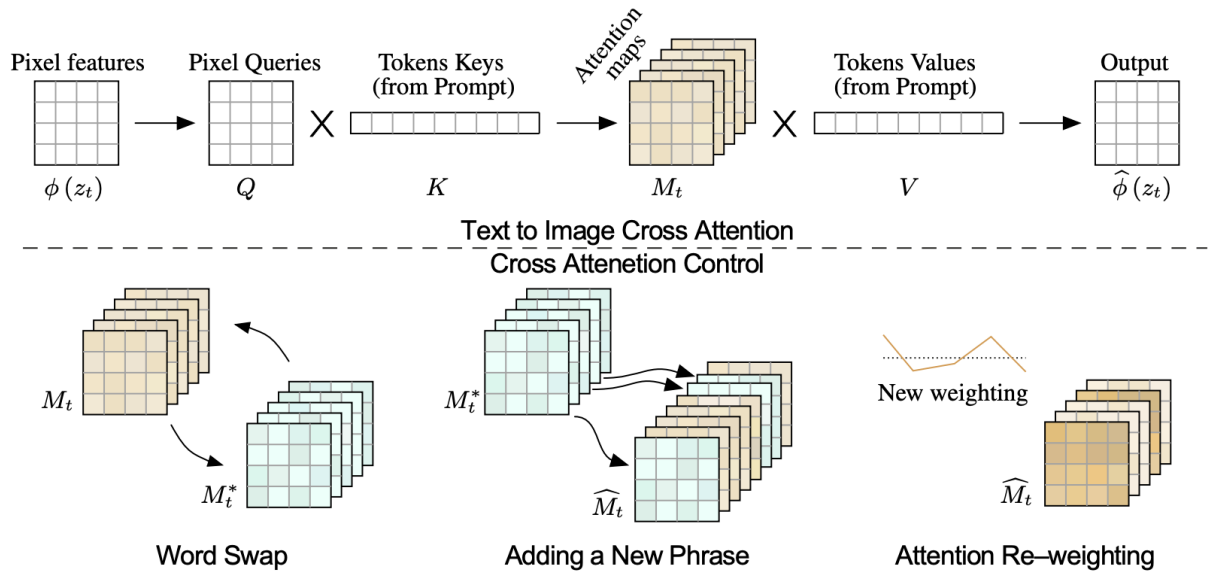
"Photo of a cat riding on a bicycle."

car



"a cake with decorations."

jelly beans



SDXL

CLIP ViT-L — the same encoder from SD1.5, producing a 768-dim per-token embedding.

OpenCLIP ViT-bigG — a much larger encoder trained on LAION-2B, producing a 1280-dim per-token embedding. It also outputs a pooled embedding.

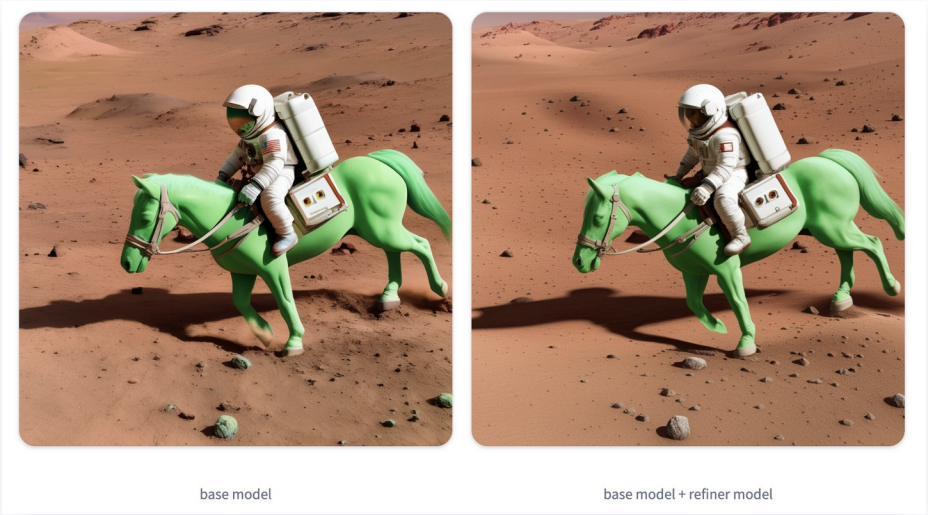
The two per-token embeddings are concatenated along the channel dimension. The pooled embedding from OpenCLIP ViT-bigG gets projected and **added directly to the timestep embedding**.

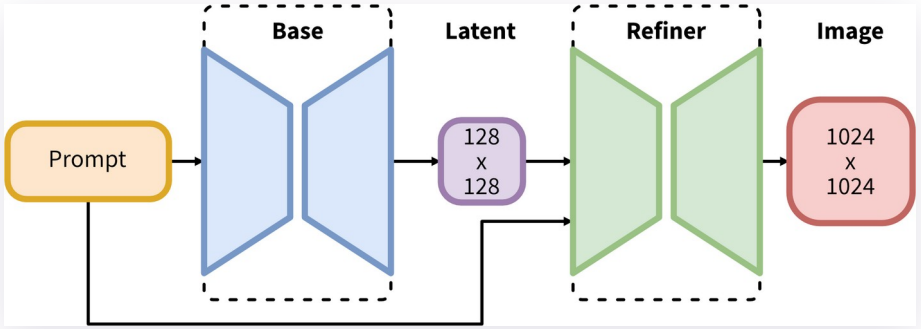
Multi-aspect training

Can generate images not only squared but also rectangle

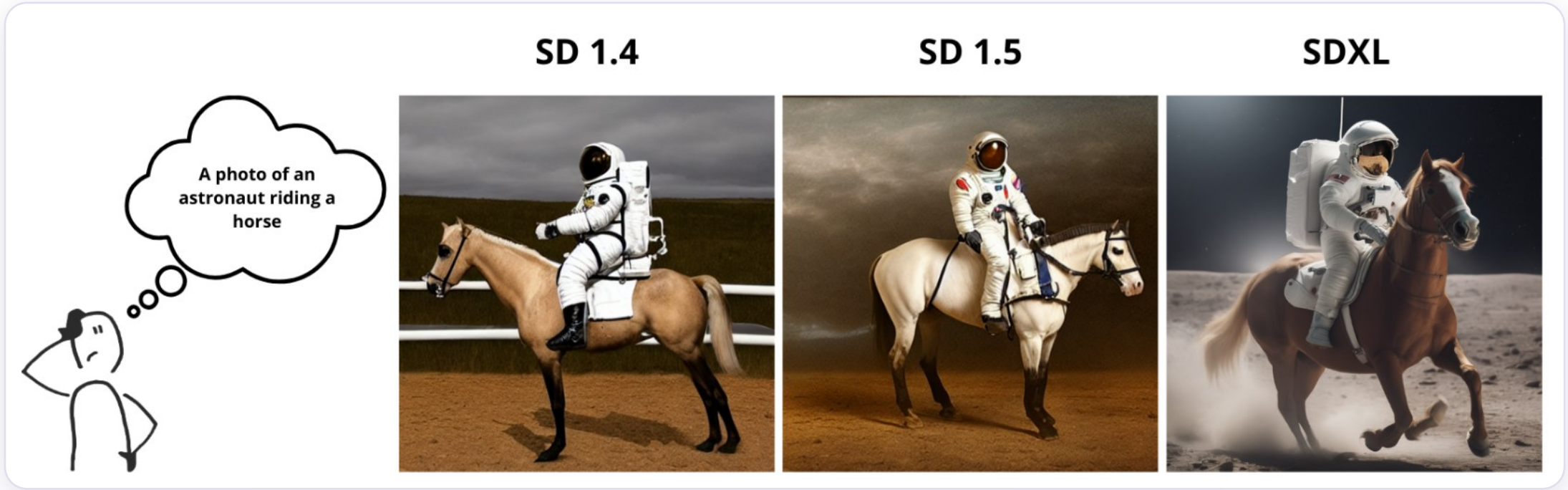
Refiner model

SDXL introduces two U-Net models — the base model and a refiner that functions as an image-to-image model





Stable diffusion UNet-based history



DLS

04

Diffusion transformers

DiT, adaLN and scaling



Diffusion transformers

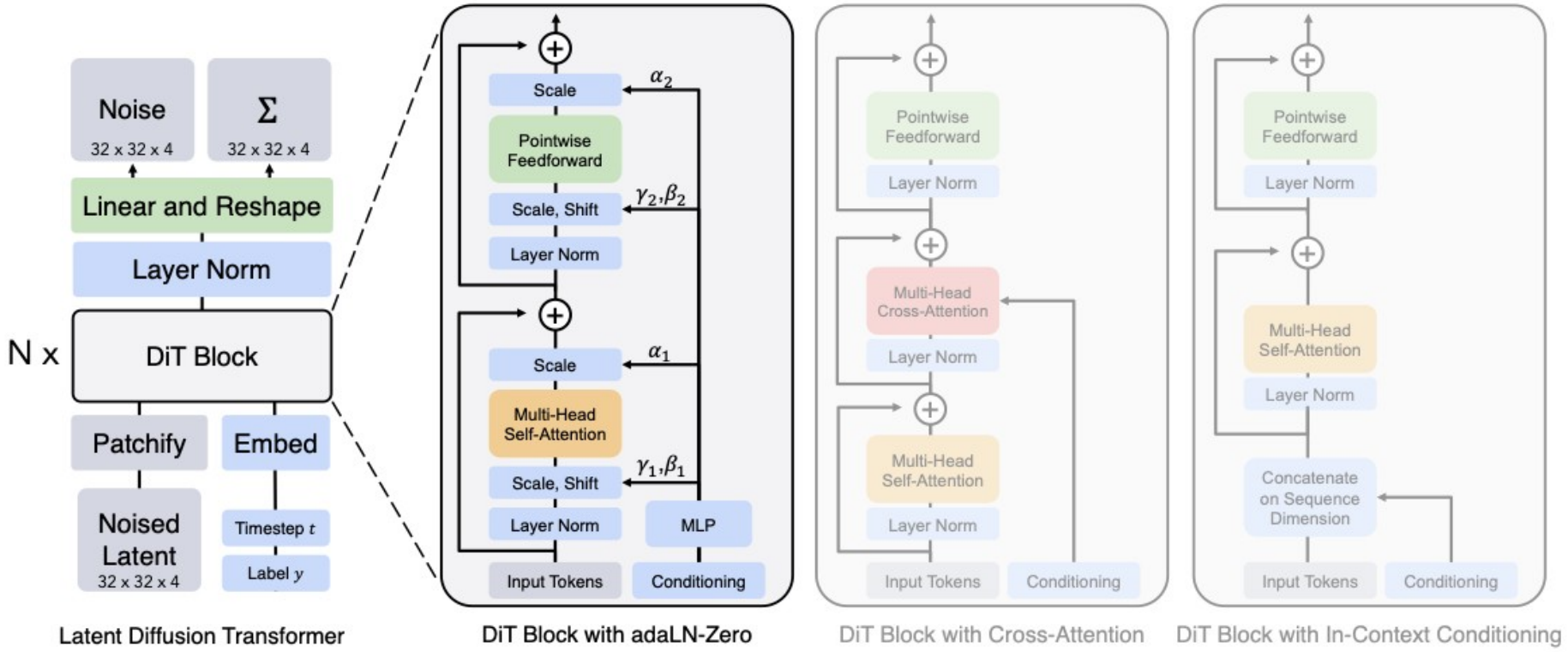
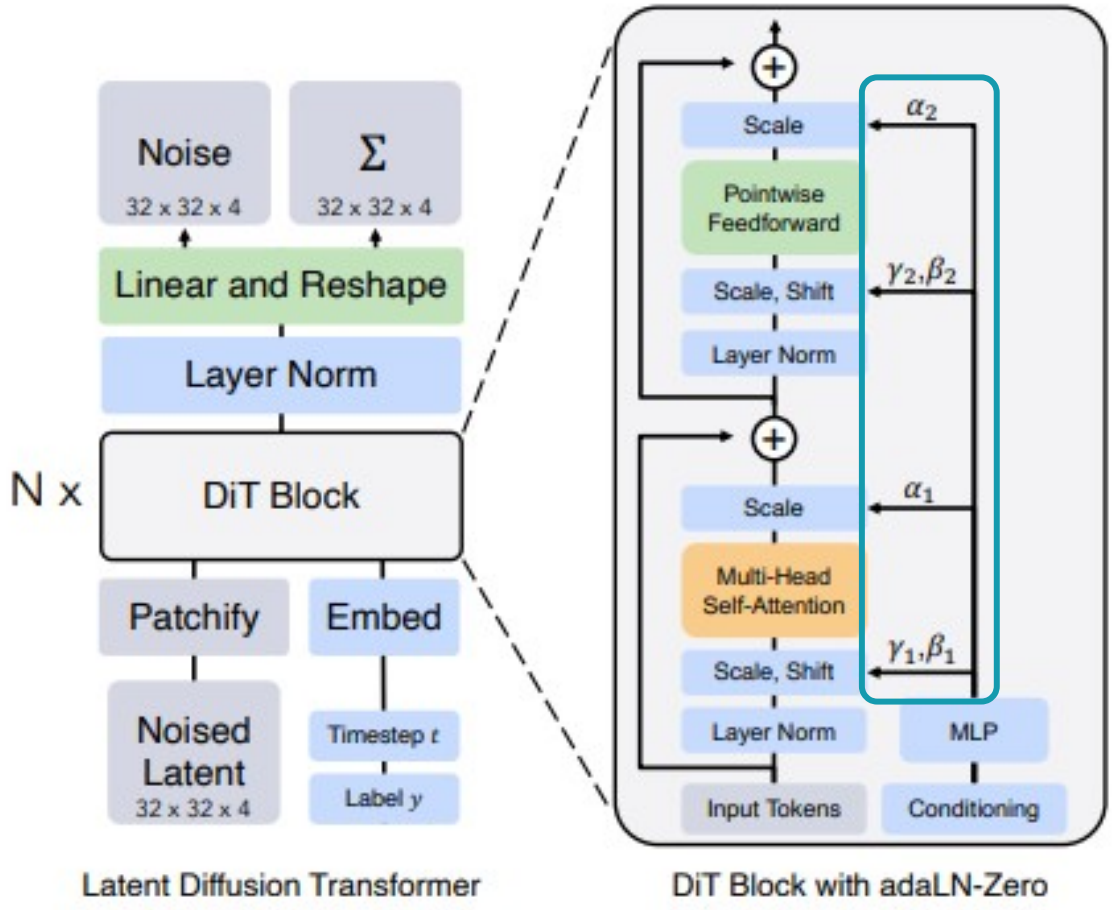


Figure 3. **The Diffusion Transformer (DiT) architecture.** *Left:* We train conditional latent DiT models. The input latent is decomposed into patches and processed by several DiT blocks. *Right:* Details of our DiT blocks. We experiment with variants of standard transformer blocks that incorporate conditioning via adaptive layer norm, cross-attention and extra input tokens. Adaptive layer norm works best.

paper →

Diffusion transformers: adaLN

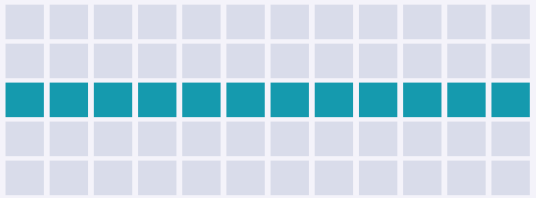


LAYER NORM, PER TOKEN

$$\mu_{b,n} = \frac{1}{D}\sum_{d=1}^{D} x_{b,n,d}$$

$$\sigma^2_{b,n} = \frac{1}{D}\sum_{d=1}^{D}(x_{b,n,d} - \mu_{b,n})^2$$

$$\text{adaLN}(x, c) = \frac{x - \mu}{\sqrt{\sigma^2 + \epsilon}} \cdot \gamma(c) + \beta(c)$$

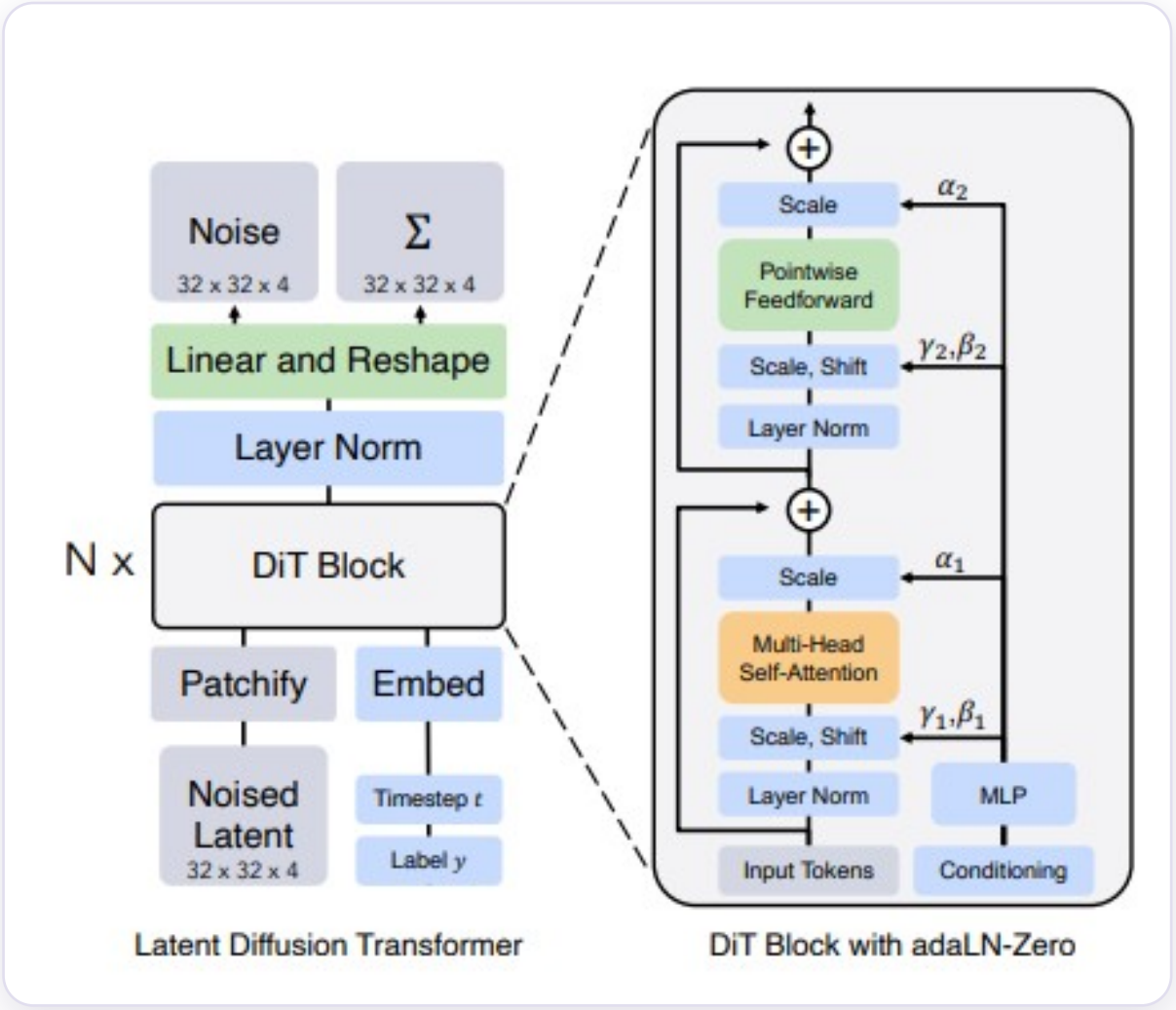


N tokens × D channels
μ, σ over the D channels of each token

Takeaway
The condition (timestep + class) only predicts γ and β.

paper →

Diffusion transformers: adaLN-Zero



$$\mathrm{AdaLN}(x, c) = \gamma(c) \cdot \mathrm{LayerNorm}(x) + \beta(c)$$

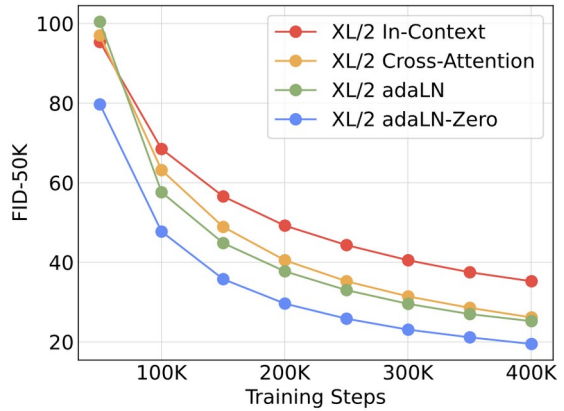


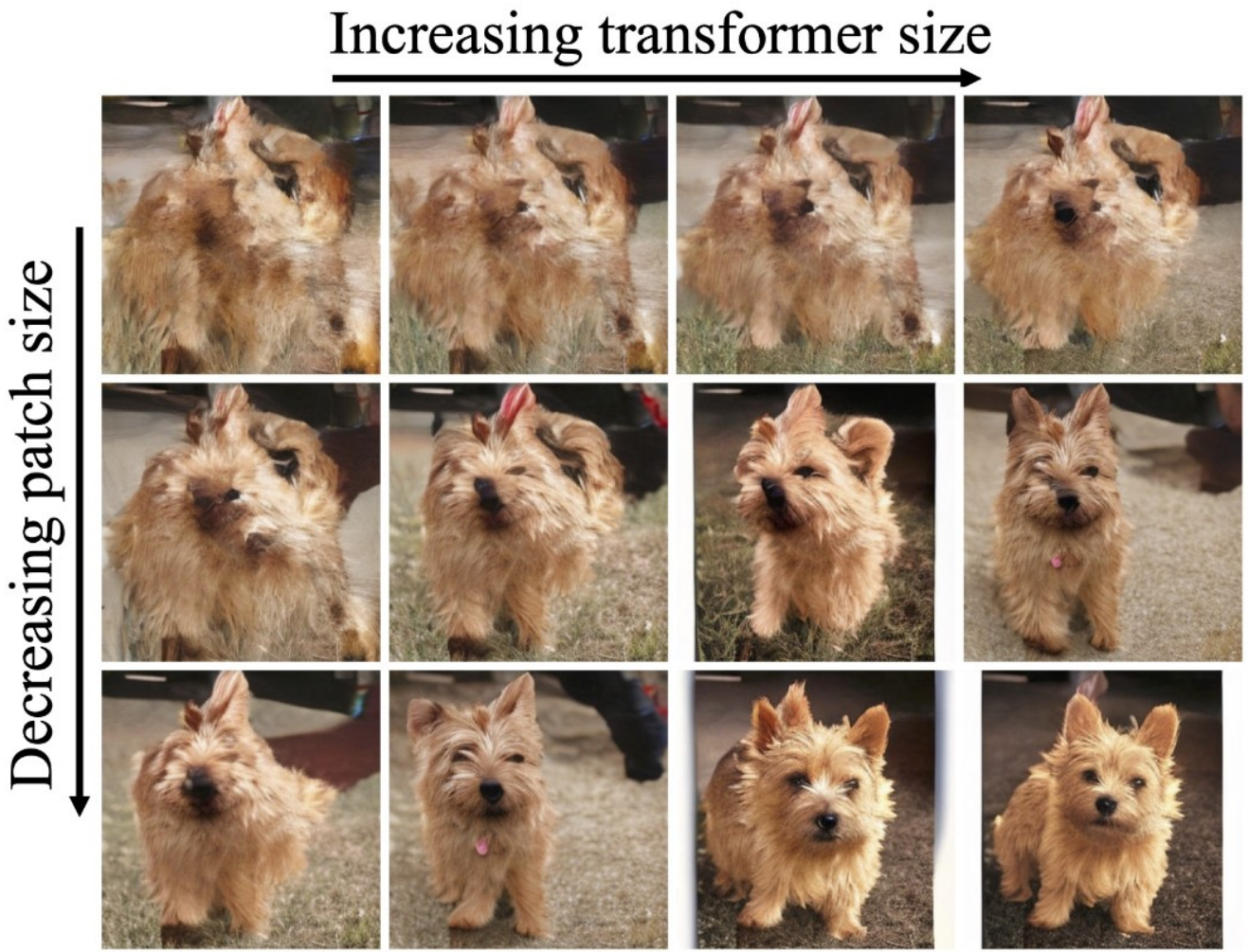
Figure 5. **Comparing different conditioning strategies.** adaLN-Zero outperforms cross-attention and in-context conditioning at all stages of training.

Takeaway

- The MLP that predicts γ, β, α is zero-initialized,
- Best FID among in-context, cross-attention and adaLN conditioning

paper →

Diffusion transformers: scaling



Two axes of scaling

- Bigger transformer: S → B → L → XL
- Smaller patches: more tokens for the same image

Takeaway

Sample quality improves along both axes: FID tracks model compute (Gflops), not just the number of parameters

DLS

05

Stable Diffusion 3

Flow matching, MM-DiT and joint attention



Stable Diffusion 3



Figure 1. High-resolution samples from our 8B rectified flow model, showcasing its capabilities in typography, precise prompt following and spatial reasoning, attention to fine details, and high image quality across a wide variety of styles.

paper →

DLS

From diffusion to flow matching (preview)

1 Straight path

interpolate data and noise

$$x_t = (1 - t)\, x_0 + t\, \epsilon$$

2 Velocity field

the network predicts the direction of motion

$$\frac{dx_t}{dt} = v_\theta(x_t, t)$$

3 Conditional flow matching loss

regress the target velocity

$$\mathcal{L}_{CFM} = \mathbb{E}_{t,\, x_0,\, \epsilon} \left\| v_\theta(x_t, t) - u_t \right\|^2$$

4 Target velocity

constant along the straight line

$$u_t = \frac{dx_t}{dt} = \epsilon - x_0$$

x_0 — data, ε — noise, $t \in [0, 1]$

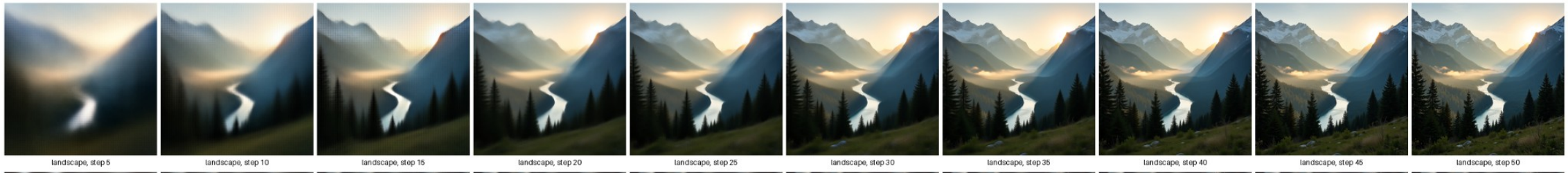
variant	ImageNet CLIP	ImageNet FID	CC12M CLIP	CC12M FID
rf	0.247	49.70	0.217	94.90
edm(-1.20, 1.20)	0.236	63.12	0.200	116.60
eps/linear	0.245	48.42	0.222	*90.34*
v/cos	0.244	50.74	0.209	97.87
v/linear	0.246	51.68	0.217	100.76
rf/lognorm(0.50, 0.60)	**0.256**	80.41	<u>0.233</u>	120.84
rf/mode(1.75)	*0.253*	**44.39**	0.218	94.06
rf/lognorm(1.00, 0.60)	<u>0.254</u>	114.26	**0.234**	147.69
rf/lognorm(-0.50, 1.00)	0.248	<u>45.64</u>	0.219	**89.70**
rf/lognorm(0.00, 1.00)	0.250	*45.78*	*0.224*	<u>89.91</u>

Table 2. **Metrics for different variants.** FID and CLIP scores of different variants with 25 sampling steps. We highlight the **best**, <u>second best</u>, and *third best* entries.

paper →

Coarse-to-fine generation

Generation goes from coarse to fine: at t ≈ 1 the model only sets a blurry layout, at t ≈ 0 it polishes details. The real decisions — composition, objects, shapes — are made at intermediate t.



t ≈ 1
relatively easy

intermediate t
difficult and important

t ≈ 0
relatively easy

paper →

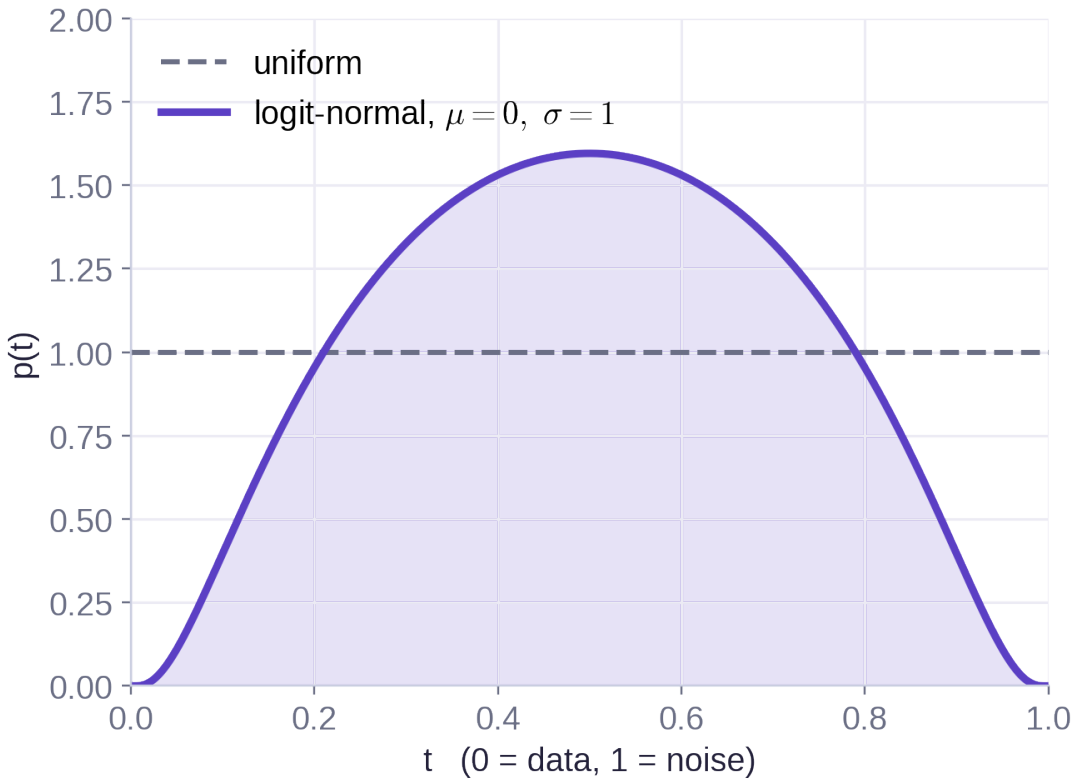
Logit-normal timestep sampling

Let’s make p(t) focus on intermediate timesteps

LOGIT-NORMAL DISTRIBUTION

$$p_{\ln}(t;\mu,\sigma) = \frac{1}{\sigma\sqrt{2\pi}\,t(1-t)}\exp\left(-\frac{(\operatorname{logit}(t)-\mu)^2}{2\sigma^2}\right)$$

$$\operatorname{logit}(t) = \log\frac{t}{1-t}, \qquad \mu = \log\alpha, \quad \sigma = 1$$



paper →

Same t, different resolution

The same noise level t destroys the image less at higher resolution

t = 0.5

same image, same t, shown at the same size

128 × 128



256 × 256



512 × 512



1024 × 1024



KEEP THE UNCERTAINTY FIXED WHEN RESOLUTION GROWS

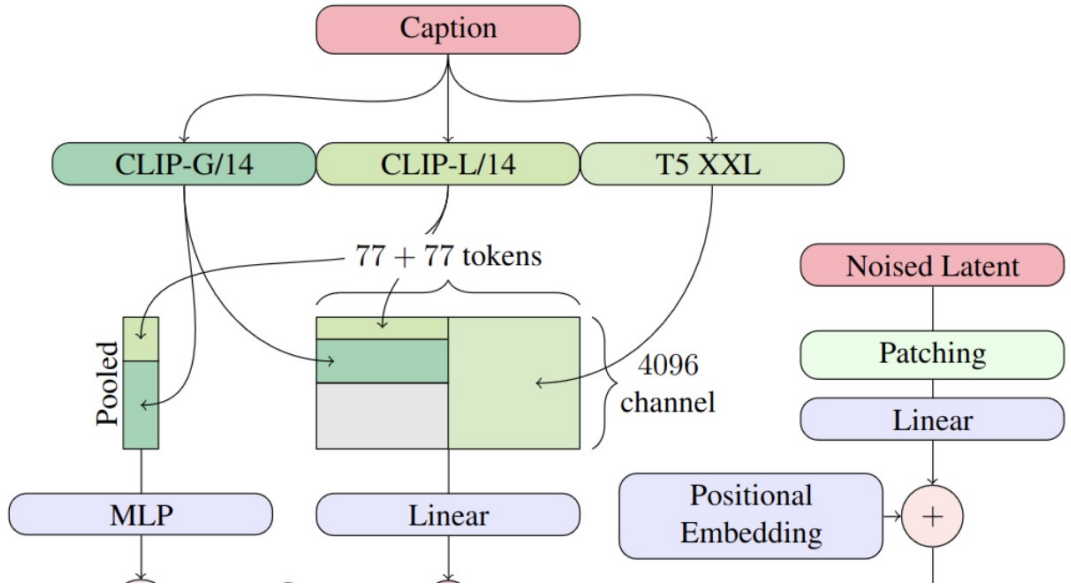
$$\sigma(t', m) = \sigma(t, n) \quad \Rightarrow \quad \frac{t'}{1-t'} = \sqrt{\frac{m}{n}} \cdot \frac{t}{1-t}$$

$$t' = \frac{\alpha\, t}{1 + (\alpha - 1)\, t}, \qquad \alpha = \sqrt{m/n}$$

This is exactly the timestep shift

m, n — pixel counts at the new and base resolution. Same as moving the logit-normal by log α

Stable Diffusion 3

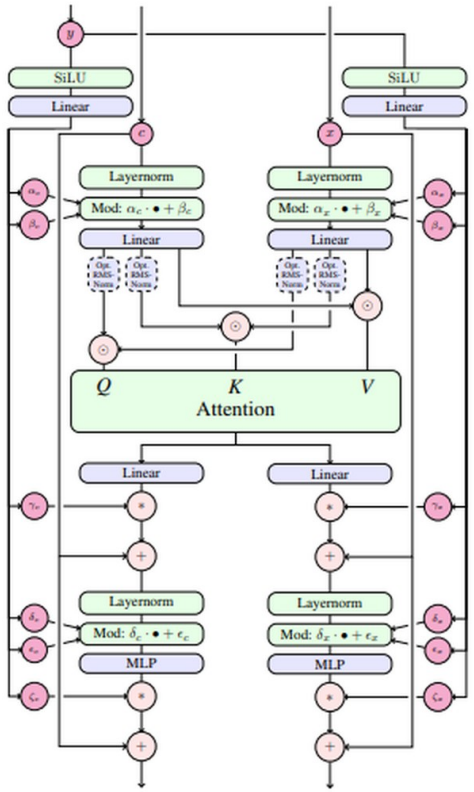


Text embeddings are used as follows:

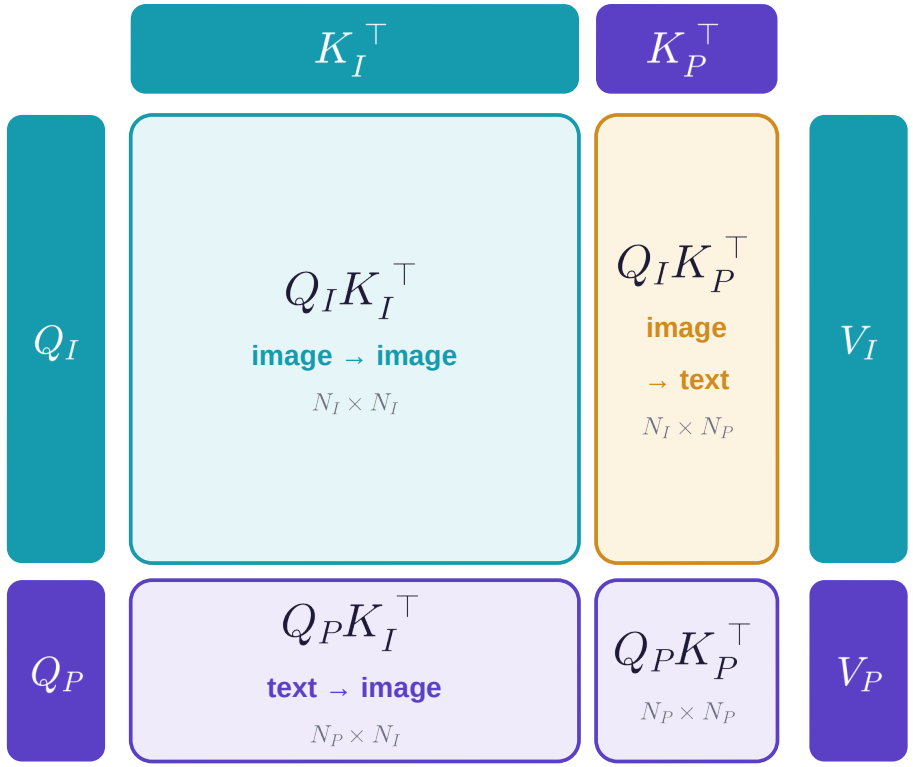
- The pooled CLIP embeddings are added to the timestep embedding and used in AdaLN
- CLIP + T5 embeddings are concatenated and fed into the MM-DiT block

paper →

Joint attention (MM-DiT)



One MM-DiT block: two streams, shared attention



self-attention: already in SD 1.x

cross-attention: already in SD 1.x

new: text tokens attend to image and text

One attention for both

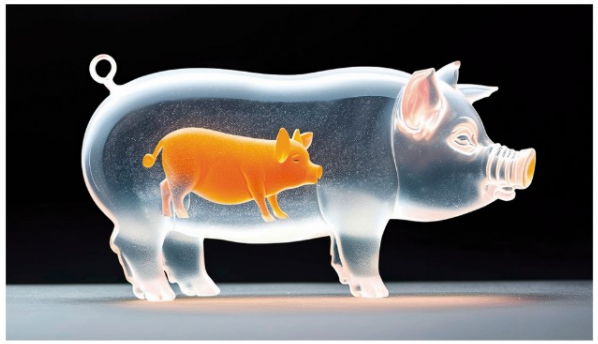
Image and text tokens are concatenated into one sequence: every query sees all image and text keys

Separate weights

Each modality keeps its own LayerNorm, adaLN modulation, QKV projections and MLP. Only the attention is shared

Stable Diffusion 3, results





Prompt: Translucent pig, inside is a smaller pig.



Prompt: A massive alien space ship that is shaped like a pretzel.



Prompt: A kangaroo holding a beer, wearing ski goggles and passionately singing silly songs.



Prompt: An entire universe inside a bottle sitting on the shelf at walmart on sale.



Prompt: A cheeseburger with juicy beef patties and melted cheese sits on top of a toilet that looks like a throne and stands in the middle of the royal chamber.



Prompt: This dreamlike digital art captures a vibrant, kaleidoscopic bird in a lush rainforest



Prompt: A car made out of vegetables.



Prompt: Heat death of the universe line art

But...

blue sphere on right, green cylinder on left



blue sphere on right with text "SD3", green cylinder on left with red cube on top



blue sphere on right with text "SD3", green cylinder on left with red cube on top, orange background, dog face at the bottom



man sitting on beach



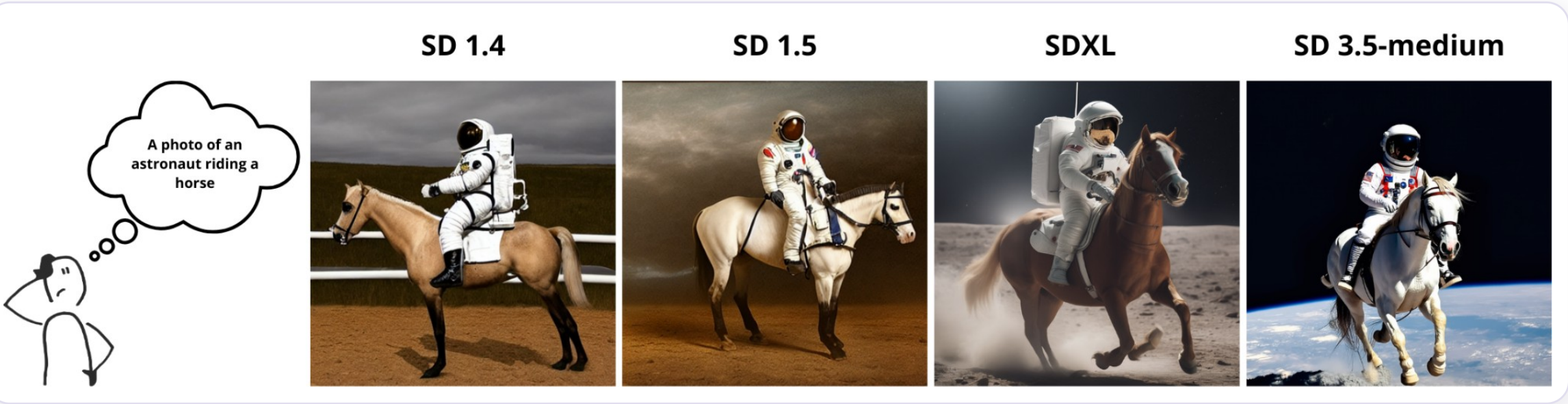








Stable diffusion history



DLS

06

FLUX

FLUX.1, FLUX Kontext and FLUX.2



code →

FLUX.1

Rectified-flow transformer, 12B parameters

[pro] closed, API only
[dev] open weights, guidance-distilled
[schnell] open, fast: 1–4 steps

WAIT… SAME PEOPLE?

R. Rombach · A. Blattmann · P. Esser

LDM 2022 → SD3 2024 → **FLUX.1** Aug 2024

left Stability AI and founded Black Forest Labs

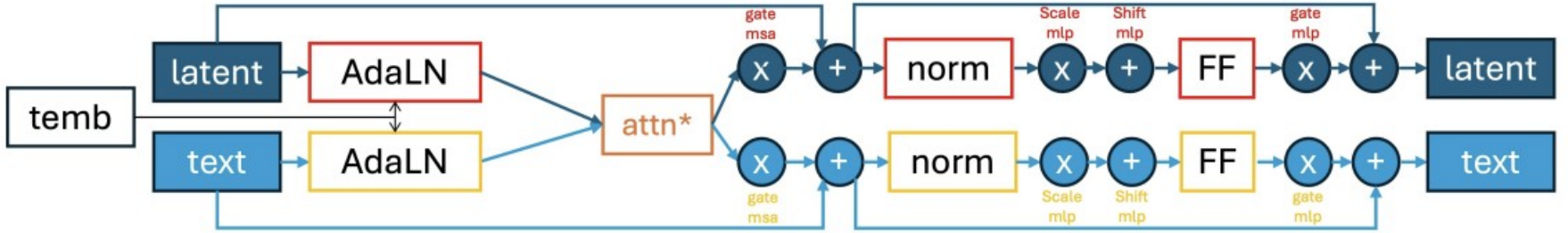




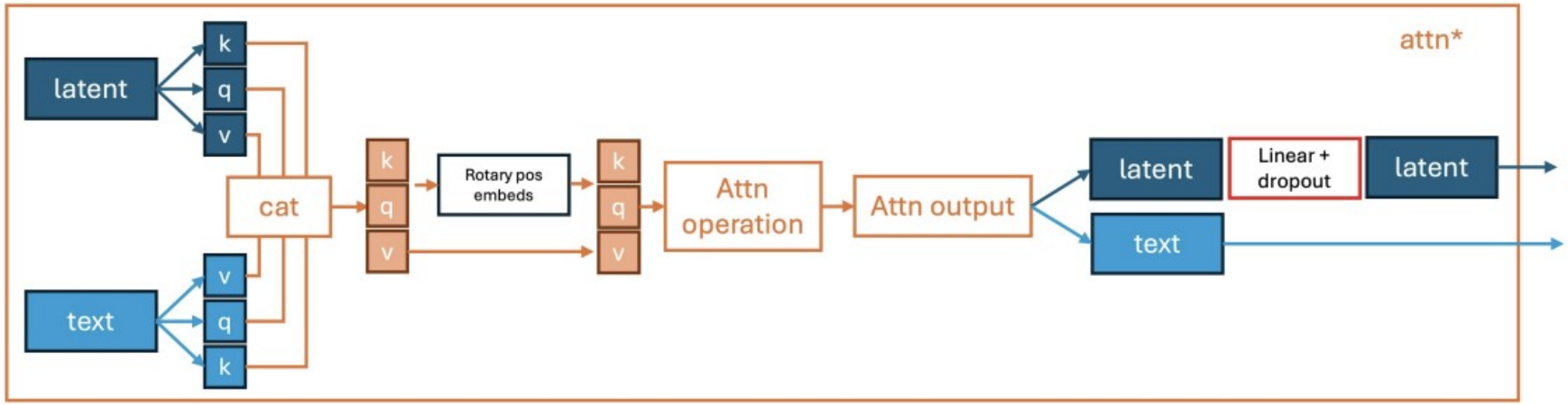
Same team, next step: the authors of Latent Diffusion and SD3 built FLUX on the same recipe: rectified flow + MM-DiT-style transformer

paper →

Double-Stream Attention Block

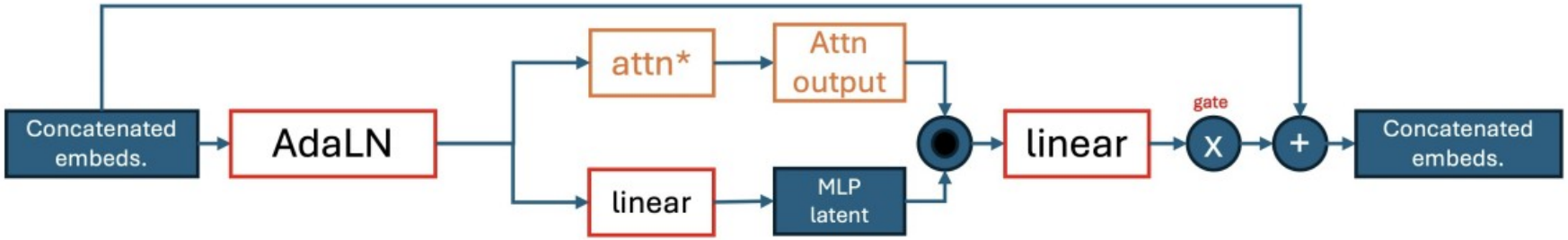


(a) Double-Stream Attention Block: latent and prompt embeddings are processed separately, in a standard multi-modal attention scheme.

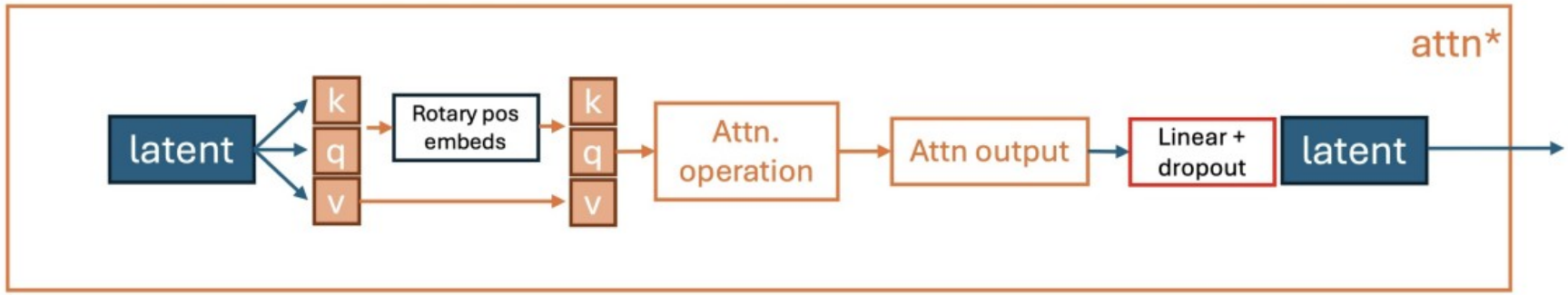


paper →

Single-Stream Attention Block



(a) Single-Stream Attention Block: latent and prompt embeddings are processed together, in a simultaneous attention scheme.

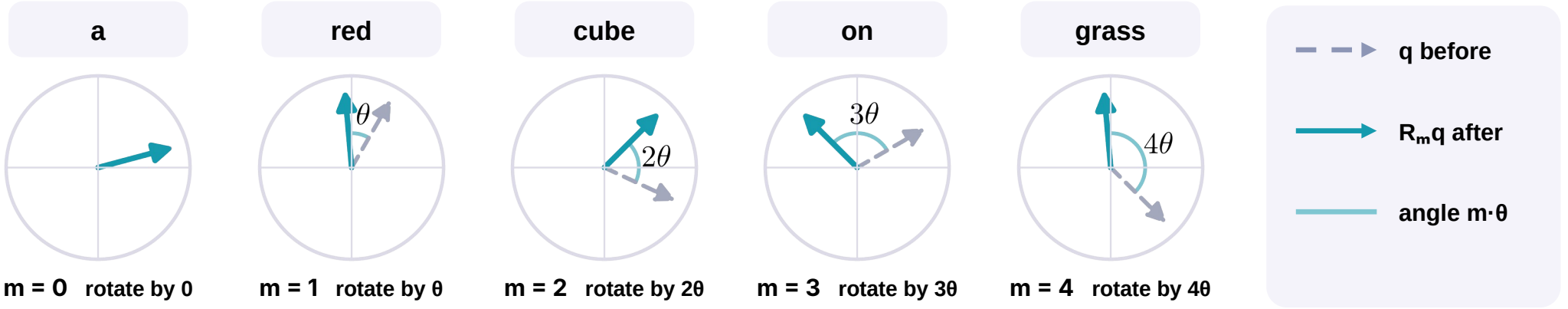


(b) The K, Q and V metrices are computed directly from the concatenated representation.

paper →

RoPE: every token is rotated

Q AND K OF THE TOKEN AT POSITION M ARE ROTATED BY THE ANGLE M·Θ



ONE PAIR OF DIMENSIONS

$$\begin{bmatrix} x'_{2i} \\ x'_{2i+1} \end{bmatrix} = \begin{bmatrix} \cos m\theta_i & -\sin m\theta_i \\ \sin m\theta_i & \cos m\theta_i \end{bmatrix} \begin{bmatrix} x_{2i} \\ x_{2i+1} \end{bmatrix}$$

q is split into d/2 pairs; each pair is a 2D point rotated by its own speed

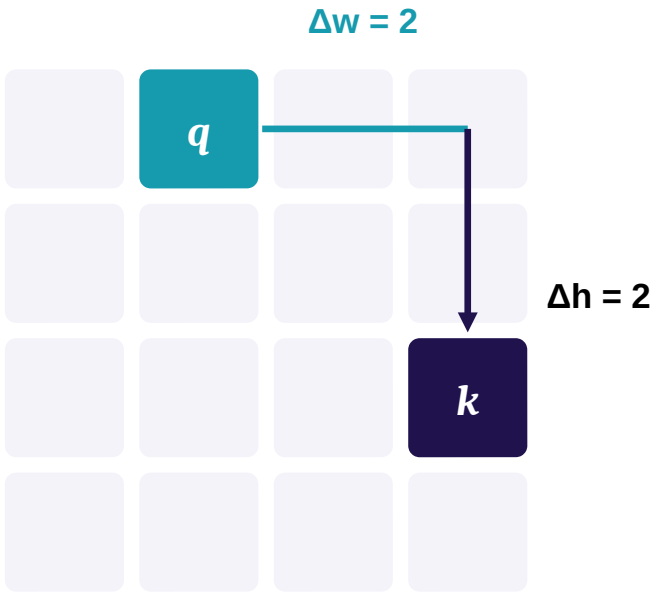
$$\theta_i = 10000^{-2i/d}$$

paper →

DLS

Axial RoPE: attention sees both offsets

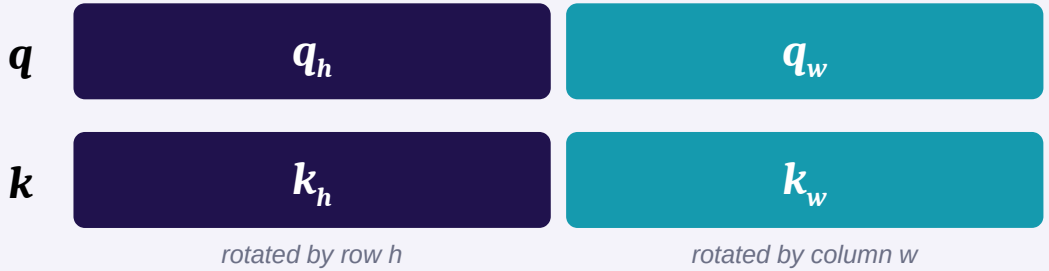
TWO TOKENS ON THE GRID



q at (0, 1) · k at (2, 3)

2 columns right and 2 rows down

EACH VECTOR IS SPLIT INTO TWO HALVES



$$\text{score} = \underbrace{\langle q_h,\ R(\Delta h)\, k_h \rangle}_{\text{rows apart}} + \underbrace{\langle q_w,\ R(\Delta w)\, k_w \rangle}_{\text{columns apart}}$$

The two halves add up: attention takes both the vertical and the horizontal offset into account

DLS

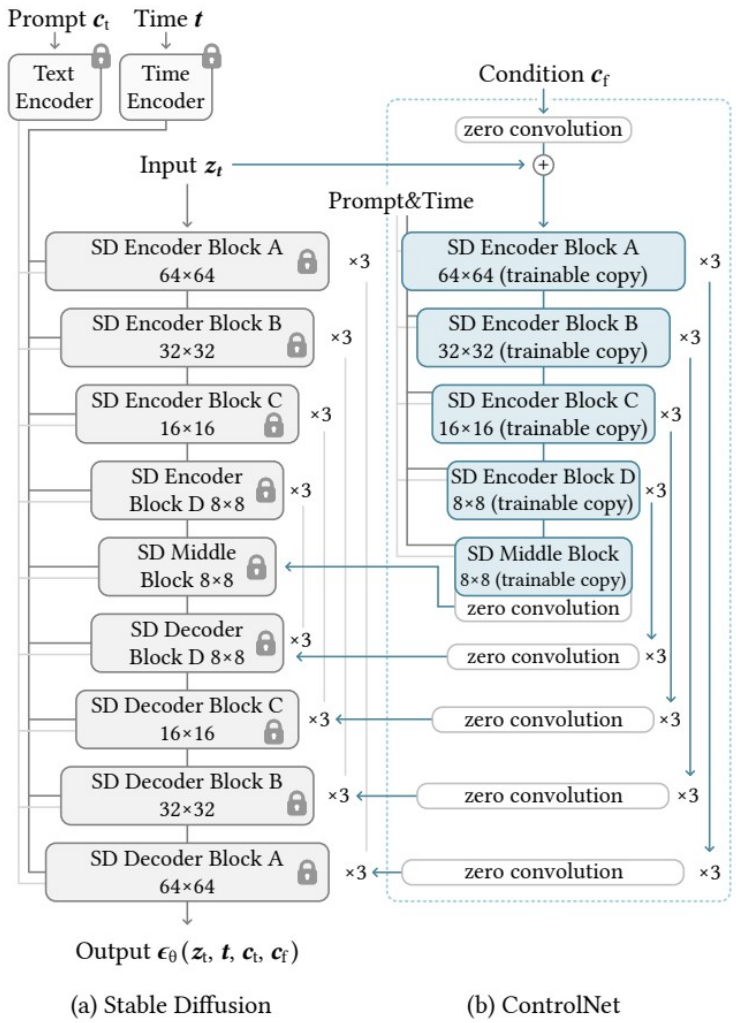
06

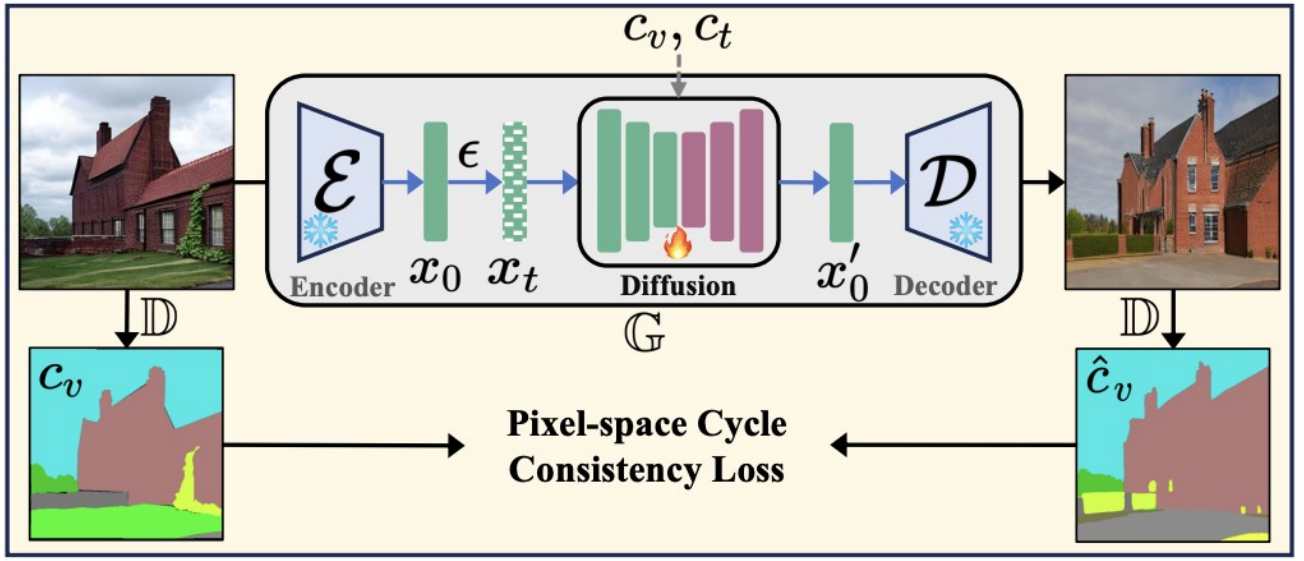
In-context models



paper → paper → paper →

ControlNet and others





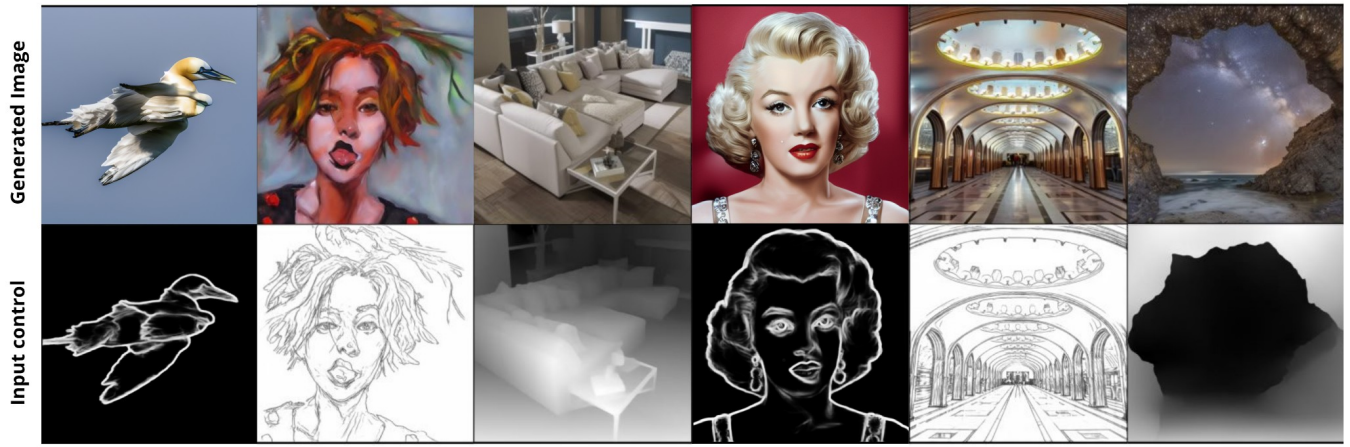


IMAGE EDITING

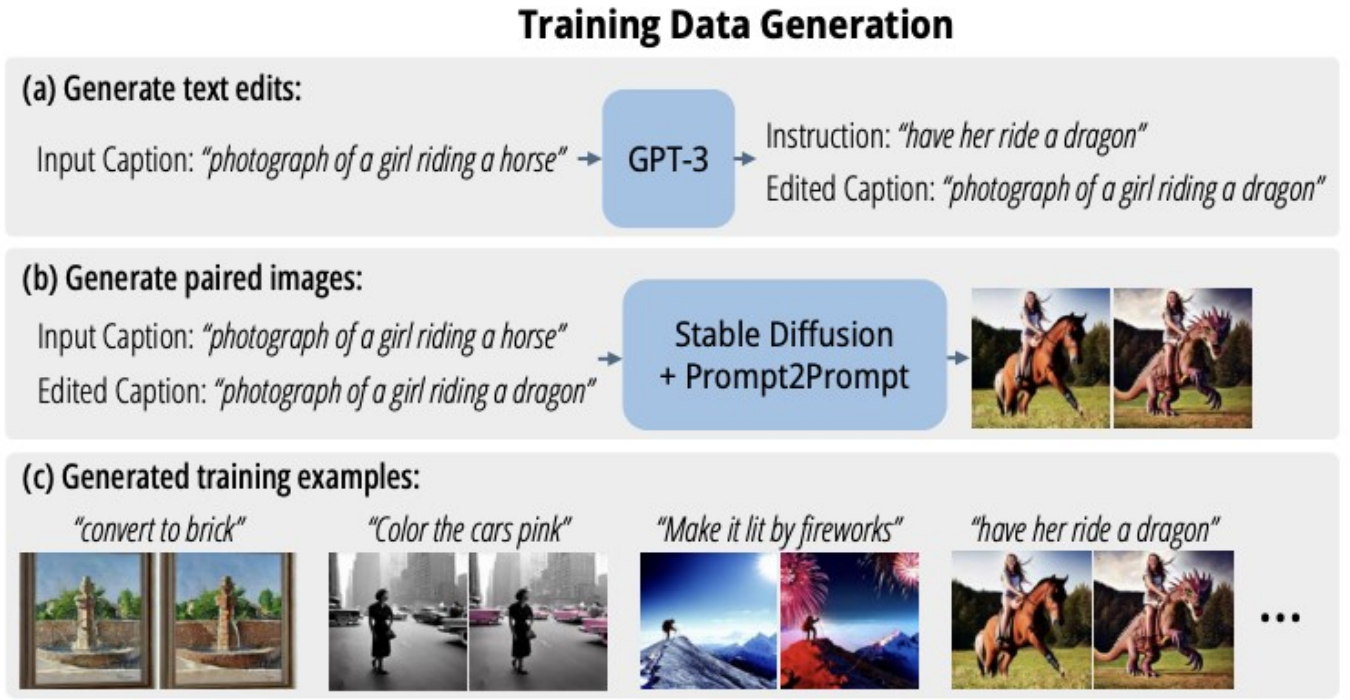
paper →

InstructPix2Pix

1. **Synthetic training pairs**
   GPT-3 writes edits, SD + Prompt-to-Prompt draws before / after

2. **Image as extra input channels**
   input latent concatenated to the noisy latent

3. **Two guidance scales**
   s_I keeps the input image, s_T pushes the edit





paper →

FLUX.1 Kontext: text + reference images

GOAL

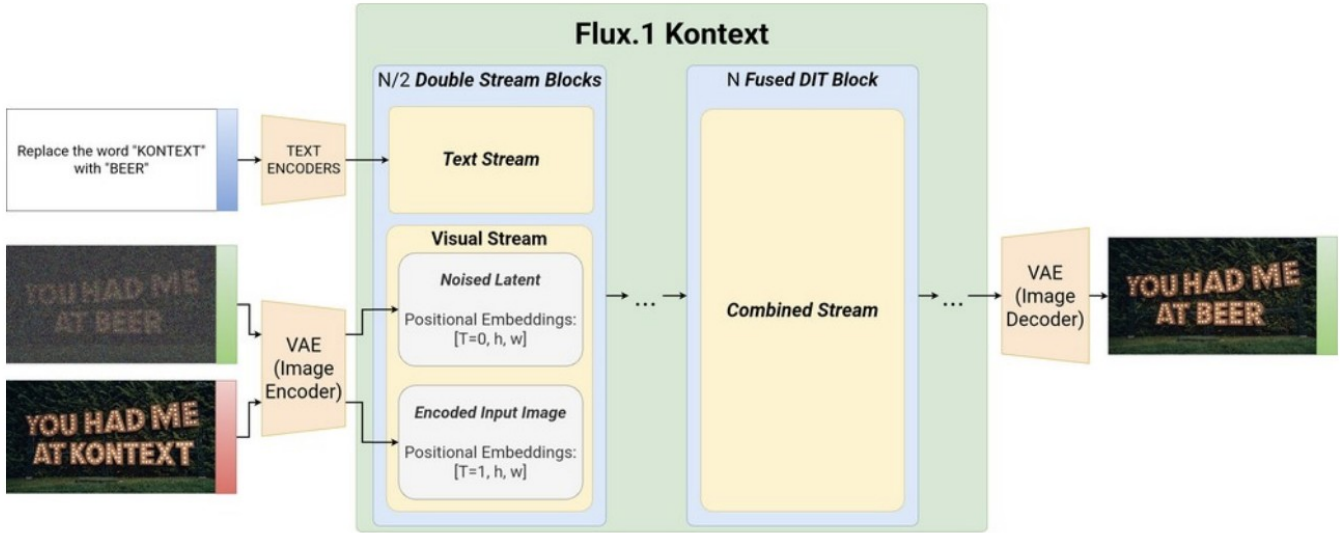
$$p_\theta(x \mid y,\ c)$$

x — target image, y — reference image(s) or ∅, c — text instruction

Key idea: sequence concatenation

Reference image → frozen VAE → tokens appended to the noisy target tokens in the visual stream.

Start: FLUX.1 text-to-image checkpoint, then joint fine-tuning on image-to-image and text-to-image



local editing
change one object, keep the rest

in-context generation
same character / style in a new scene

text-to-image
no reference: y = ∅

paper →

FLUX.1 Kontext: 3D RoPE

Every image token gets a position **(t, h, w)** — t tells the images apart, (h, w) is the place inside the image

target (noisy) t = 0

0,0	0,1	0,2	0,3
1,0	1,1	1,2	1,3
2,0	2,1	2,2	2,3
3,0	3,1	3,2	3,3

reference 1 t = 1

0,0	0,1	0,2	0,3
1,0	1,1	1,2	1,3
2,0	2,1	2,2	2,3
3,0	3,1	3,2	3,3

reference 2 t = 2

0,0	0,1	0,2	0,3
1,0	1,1	1,2	1,3
2,0	2,1	2,2	2,3
3,0	3,1	3,2	3,3

same (h, w) grid in every image · only the index t changes

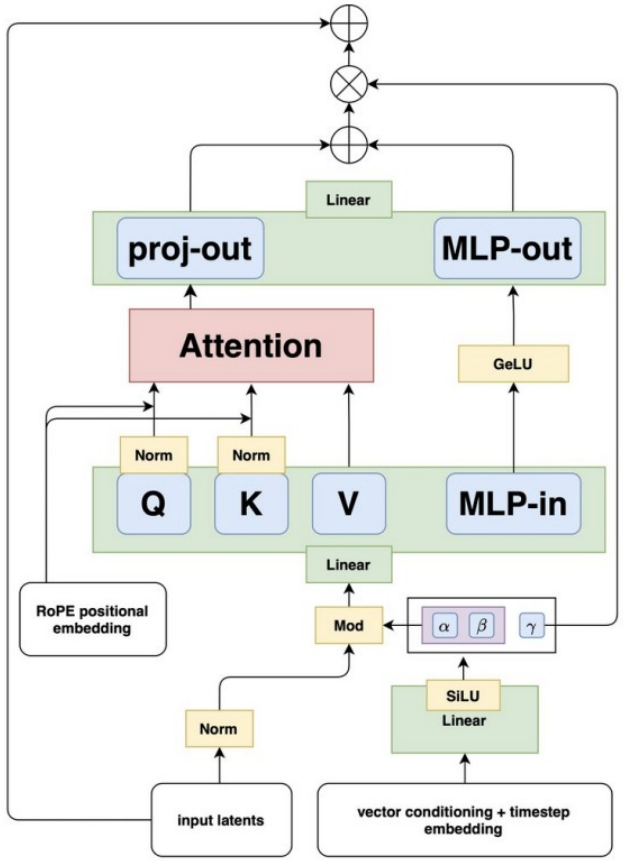


Figure 3: A fused DiT block equipped with rotary positional embeddings

FLUX Kontext generation (Results)



(a) Input image



(b) *"extract only the skirt over a white background, product photography style"*



(c) *"show me an extreme closeup of the fabric"*

Figure 6: **Product Photography.** (a) Input image showing the full outfit. (b) Extracted skirt on a white background in a product-photography style. (c) Extreme close-up of the skirt's fabric, highlighting texture and pattern details.



(a) Input image



(b) *"Using this style, a kid on a bicycles rolls through desert ruins, spotlights scanning ancient scrolls projected as holographic sandstorms."*



(c) *"Using this style, a grand piano made of shifting mirrors performs itself for an audience of empty velvet chairs in zero-gravity."*



(e) Input image



(f) *"Using this style, a half-folded metropolis hangs from steel strings over an ink-wash ocean while cranes of light sketch new streets in mid-air."*



(g) *"Using this style, a dapper octopus conducts a jazz duo of owls on a shimmering moonlit bandstand."*

code →

DLS

FLUX.2: architecture and model family

ARCHITECTURE

Mistral-3 · 24B VLM
reads the prompt and the reference images

Rectified flow transformer
32B in [dev]

new VAE · better reconstruction

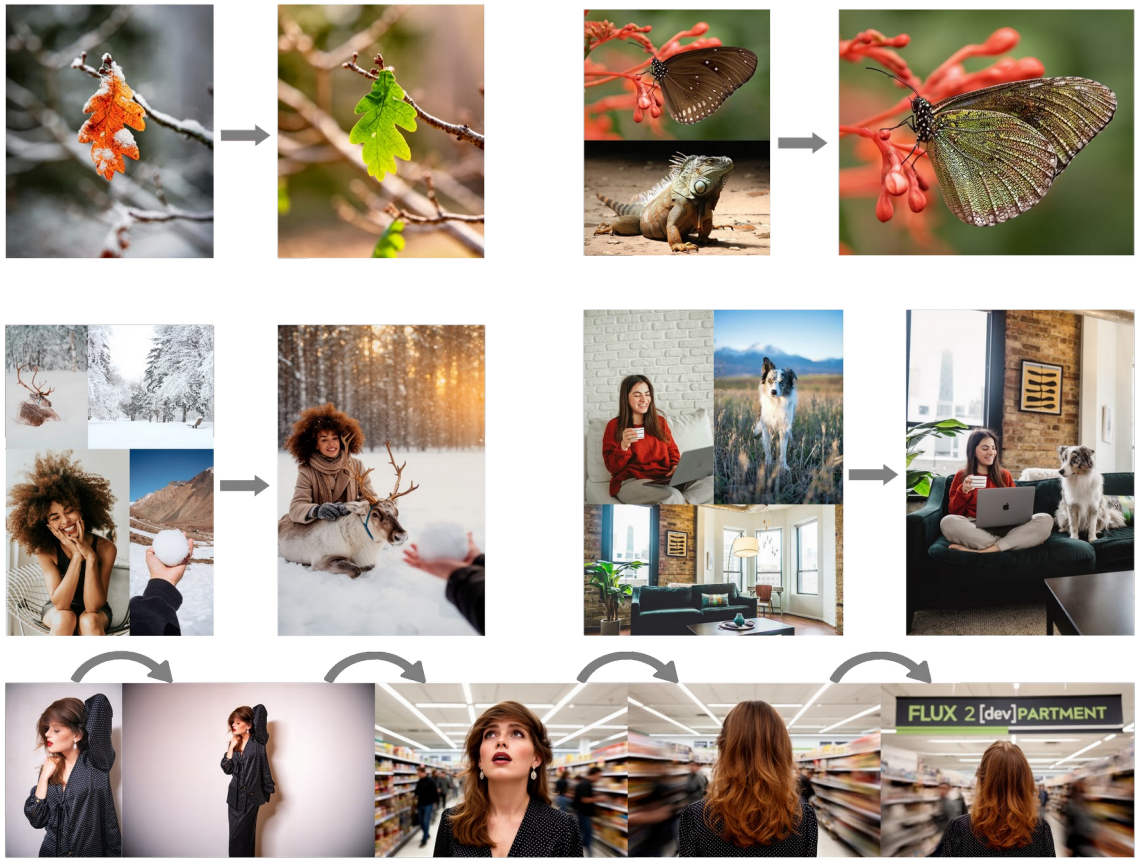
MODEL FAMILY

[pro]
closed · production

[flex]
closed · you set steps and guidance

[dev]
open weights · 32B

[klein]
distilled · 9B and 4B



But...

A photo of a horse riding an astronaut



DLS

07

Open-weight models

Leaderboards



Text-to-image arena

Summer 2026

	Range	Creator	Model
1	1–3	Fal	FLUX.2 [dev] Turbo · Open Weights
2	1–3	Black Forest Labs	FLUX.2 [dev] · Open Weights
3	1–3	Alibaba	Qwen Image Max 2512 · Open Weights
4	4	Fal	FLUX.2 [dev] Flash · Open Weights
5	5–6	Tencent	HunyuanImage 3.0 (Fal) · Open Weights
6	5–6	Black Forest Labs	FLUX.2 [klein] 9B · Open Weights
7	7–8	Tencent	HunyuanImage 3.0 Instruct (Fal) · Open Weights
8	7–8	Alibaba	Z-Image Turbo · Open Weights
9	9	Black Forest Labs	FLUX.2 [klein] Base 9B · Open Weights

October 2026

	Range	Model
1	1–2	Qwen-Image-2.1
2	1–2	Fibo 2.0
3	3	Ideogram 4.0 (Quality)
4	4–8	Ideogram 4.0
5	5	FLUX.2 [dev]
6	4–8	Ming-Image-0.1-Design
7	4–11	FLUX.2 [dev] Turbo
8	4–11	Cosmos3-Super-Text2Image (ag...
9	6–13	FLUX.2 [dev] Flash

artificialanalysis.ai/image/leaderboard/text…